FACULTY OF ENGINEERING • KHON KAEN UNIVERSITY

Fundamental Electrical Engineering (FunE)

Final Exam Master Flashcards & Solved Problems Booklet (72 Cards)

COURSE & INSTRUCTOR
Assoc. Prof. Nararat Ruangchaijatupon

COVERAGE
Chapters 6, 7, 8, 9, 10 & Semiconductors

CONTENT INCLUSIONS
72 Cards • 22 Circuit Diagrams • Step-by-Step Solutions

PART 1

Chapter 6: Three-Phase Circuits

Balanced & Unbalanced Y-Y, Y-Δ, Complex Power, Neutral Shift, 2-Wattmeter Method (10 Flashcards)

CH06-01 Chapter 6 Level 1 (Core Concept) 📌 Lecture Slides & Formula Sheet

Star (Y) vs Delta (Δ) Voltage & Current Relations

QUESTION / EXAM PROBLEM:

In a balanced 3-phase positive-sequence (abc) system, state the exact relationships between line quantities (V_L, I_L) and phase quantities (V_p, I_p) for both Y-connection and Δ-connection.

Key Formula / Operating Principle:
Y: $V_L = \sqrt{3}V_p\angle +30^\circ, I_L = I_p$
Δ: $V_L = V_p, I_L = \sqrt{3}I_p\angle -30^\circ$

Step-by-Step Solution & Calculation:

1. Star (Y) Connection:

- Line voltage leads phase voltage by 30°: $V_{ab} = \sqrt{3}V_{an}\angle +30^\circ$
- Line current equals phase current: $I_L = I_p$

2. Delta (Δ) Connection:

- Line voltage equals phase voltage: $V_L = V_p$
- Line current lags phase current by 30°: $I_a = \sqrt{3}I_{ab}\angle -30^\circ$

⚠️ Exam Pitfall / Pro Tip:In negative sequence (acb), line voltage leads by -30° (or lags by 30°). Remember: in Y, currents are identical; in Δ, voltages are identical.

CH06-02 Chapter 6 Level 1 (Core Concept) 📌 HW Ch.6 Problem 1 & 5

Delta to Wye (Δ ↔ Y) Impedance Conversion

QUESTION / EXAM PROBLEM:

If a balanced 3ϕ load is connected in Δ with impedance $Z_\Delta = 60 + j30\ \Omega$, what is the equivalent impedance per phase when converted to a Wye (Y) connection (Z_Y)?

Key Formula / Operating Principle:
$Z_Y = \frac{Z_\Delta}{3}$

Step-by-Step Solution & Calculation:

$$Z_Y = \frac{Z_\Delta}{3} = \frac{60 + j30}{3} = 20 + j10\ \Omega$$

Conversely, if converting from Wye to Delta: $Z_\Delta = 3Z_Y$.

⚠️ Exam Pitfall / Pro Tip:Multiplying by 3 instead of dividing is a classic mistake! A Delta load draws 3 times more current than a Wye load of the same impedance value, so equivalent Wye impedance must be 3 times smaller.

CH06-03 Chapter 6 Level 1 (Core Concept) 📌 Formula Sheet & Lecture Slides

Three-Phase Power Formulas & Complex Power

QUESTION / EXAM PROBLEM:

State the formulas for balanced 3ϕ Real Power ($P_{3\phi}$), Reactive Power ($Q_{3\phi}$), Apparent Power ($|S_{3\phi}|$), and Complex Power ($S_{3\phi}$) in terms of line values (V_L, I_L).

Key Formula / Operating Principle:
$P_{3\phi} = \sqrt{3} V_L I_L \cos\theta \quad [\text{W}]$
$Q_{3\phi} = \sqrt{3} V_L I_L \sin\theta \quad [\text{VAR}]$
$|S_{3\phi}| = \sqrt{3} V_L I_L \quad [\text{VA}]$
$S_{3\phi} = \sqrt{3} V_L I_L^* = P + jQ$

Step-by-Step Solution & Calculation:
When using Line values (V_L, I_L), always include the factor $\sqrt{3}$.
When using Phase values (V_p, I_p), use factor 3:
- $P = 3V_pI_p\cos\theta$
- $Q = 3V_pI_p\sin\theta$
- $S = 3V_pI_p^*$

where $\theta = \angle Z_p$ is the load phase impedance angle, and $PF = \cos\theta$.

⚠️ Exam Pitfall / Pro Tip:Never mix $\sqrt{3}$ with phase quantities! It is either $\sqrt{3}V_LI_L$ or $3V_pI_p$.

Balanced 4-Wire Y-Y System Full Calculation

QUESTION / EXAM PROBLEM:

In the balanced Y-Y circuit shown below, the source phase voltages are $V_{an} = 220\angle 0^\circ$ V, $V_{bn} = 220\angle -120^\circ$ V, $V_{cn} = 220\angle -240^\circ$ V. Line impedance is $Z_{\text{line}} = 10 + j5\ \Omega$. Balanced load impedance is $Z_L = 20 + j50\ \Omega$. Find: (1) Line voltages V_{ab}, V_{bc}, V_{ca}, (2) Line currents I_{aA}, I_{bB}, I_{cC}, (3) Neutral current I_{Nn}, (4) Total 3ϕ complex power S.

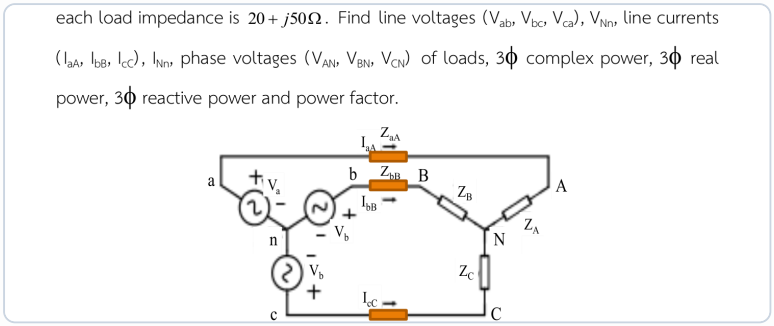


Fig. 1: Balanced Y-Y System with Line Impedance

Key Formula / Operating Principle:

$V_{ab} = \sqrt{3} V_{an} \angle +30^\circ$

$I_{aA} = \frac{V_{an}}{Z_{\text{line}} + Z_L}$

$I_{Nn} = 0$

$S_{3\phi} = 3 V_{an} I_{aA}^*$

Step-by-Step Solution & Calculation:

1. Line Voltages:

- $V_{ab} = \sqrt{3}(220)\angle(0^\circ + 30^\circ) = 381.05\angle 30^\circ$ V
- $V_{bc} = 381.05\angle -90^\circ$ V
- $V_{ca} = 381.05\angle -210^\circ$ V

2. Per-phase total impedance:

$Z_{\text{total}} = (10 + j5) + (20 + j50) = 30 + j55\ \Omega = 62.65\angle 61.39^\circ\ \Omega$

3. Line Currents:

- $I_{aA} = \frac{220\angle 0^\circ}{62.65\angle 61.39^\circ} = 3.51\angle -61.39^\circ$ A
- $I_{bB} = 3.51\angle -181.39^\circ$ A
- $I_{cC} = 3.51\angle 58.61^\circ$ A

4. Neutral Current:

In a balanced system, $I_{Nn} = -(I_{aA} + I_{bB} + I_{cC}) = 0$ A. $V_{Nn} = 0$ V.

5. Total 3ϕ Power:

$S = 3 \times 220 \times 3.51\angle +61.39^\circ = 2,316.6\angle 61.39^\circ\ \text{VA} = 1,109\ \text{W} + j2,034\ \text{VAR}.$

⚠️ Exam Pitfall / Pro Tip:In a balanced system, neutral voltage shift V_{Nn} is strictly ZERO, and neutral current $I_{Nn} = 0$.

Unbalanced 3-Wire Y-Y System (Neutral Shift)

QUESTION / EXAM PROBLEM:

In the 3-wire Y-Y system below (NO neutral wire), source is 220 V balanced, $Z_{\text{line}} = 10 + j5\ \Omega$, but load is UNBALANCED: $Z_A = 20 + j50\ \Omega$, $Z_B = 10 + j50\ \Omega$, $Z_C = 20 + j20\ \Omega$. How is the neutral voltage shift V_{Nn} calculated, and why do load phase voltages become unbalanced?

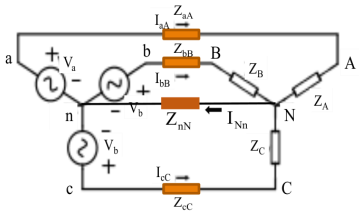


Fig. 2: Unbalanced 3-Wire Y-Y System without Neutral

Key Formula / Operating Principle:
$V_{Nn} = \frac{V_{an}Y_A + V_{bn}Y_B + V_{cn}Y_C}{Y_A + Y_B + Y_C}$ (Millman's Theorem)

Step-by-Step Solution & Calculation:

1. Millman's Theorem for Neutral Shift:

Define total branch admittance $Y_A = 1/(Z_{\text{line}} + Z_A)$, $Y_B = 1/(Z_{\text{line}} + Z_B)$, $Y_C = 1/(Z_{\text{line}} + Z_C)$.

$$V_{Nn} = \frac{V_{an}Y_A + V_{bn}Y_B + V_{cn}Y_C}{Y_A + Y_B + Y_C} \neq 0$$

2. Line Currents with Neutral Shift:

- $I_{aA} = (V_{an} - V_{Nn})Y_A$
- $I_{bB} = (V_{bn} - V_{Nn})Y_B$
- $I_{cC} = (V_{cn} - V_{Nn})Y_C$

Since no neutral wire exists, $I_{aA} + I_{bB} + I_{cC} = 0$.

3. Impact on Load Phase Voltages:

Load phase voltage $V_{AN} = V_{an} - I_{aA}Z_{\text{line}} - V_{Nn}$. Because $V_{Nn} \neq 0$, phase voltages across the load become severely distorted and unbalanced.

⚠️ Exam Pitfall / Pro Tip:Without a neutral wire, the neutral point floats ($V_{Nn} \neq 0$), causing dangerous overvoltage on lighter-loaded phases!

Unbalanced Y-Y System with Neutral Line Impedance

QUESTION / EXAM PROBLEM:

In the 4-wire Y-Y system below, an unbalanced load ($Z_A \neq Z_B \neq Z_C$) is connected with a neutral return wire having finite impedance Z_{nN}. Write the exact node-voltage equation to solve for V_{Nn} and neutral current I_{Nn}.

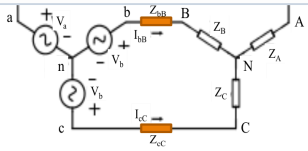


Fig. 3 Y to Y circuit with line impedance and unbalanced load.

Adapted from: https://en.wikipedia.org/wiki/Three-phase_electric_power

Fig. 3: Unbalanced 4-Wire Y-Y System with Neutral Impedance

Key Formula / Operating Principle:
$V_{Nn} = \frac{V_{an}Y_A + V_{bn}Y_B + V_{cn}Y_C}{Y_A + Y_B + Y_C + Y_n}$
$I_{Nn} = V_{Nn}Y_n = \frac{V_{Nn}}{Z_{nN}}$

Step-by-Step Solution & Calculation:

1. Node-Voltage Equation at node N:

By KCL at load neutral node N:

$$\frac{V_{Nn} - V_{an}}{Z_A + Z_{\text{line}}} + \frac{V_{Nn} - V_{bn}}{Z_B + Z_{\text{line}}} + \frac{V_{Nn} - V_{cn}}{Z_C + Z_{\text{line}}} + \frac{V_{Nn}}{Z_{nN}} = 0$$

2. Solving for V_{Nn}:

$$V_{Nn} = \frac{V_{an}Y'_A + V_{bn}Y'_B + V_{cn}Y'_C}{Y'_A + Y'_B + Y'_C + Y_n}$$

where $Y'_k = 1/(Z_k + Z_{\text{line}})$ and $Y_n = 1/Z_{nN}$.

3. Neutral Return Current:

$$I_{Nn} = \frac{V_{Nn}}{Z_{nN}} = -(I_{aA} + I_{bB} + I_{cC})$$

.

⚠️ Exam Pitfall / Pro Tip:Do not omit $Y_n = 1/Z_{nN}$ in the denominator of Millman's formula! As $Z_{nN} \to 0$, $V_{Nn} \to 0$ (solid ground).

CH06-07 Chapter 6 Level 2 (Past Exam / Calc) HW Ch.6 Problem 4 (Page 3)

Y-Connected Induction Motor Load Calculation

QUESTION / EXAM PROBLEM:

A 3ϕ, 12-hp induction motor is Y-connected to a 3ϕ supply with line-to-line voltage $V_L = 380\ \text{V}, 50\ \text{Hz}$. The motor has $\eta = 90\%$ efficiency and a lagging power factor of 0.85. Calculate: (1) Input real power (P_{in}), (2) Line current (I_L), (3) Per-phase impedance magnitude ($|Z|$), resistance (R), and inductance (L). ($1\ \text{hp} = 746\ \text{W}$)

Key Formula / Operating Principle:

$P_{\text{out}} = 12 \times 746\ \text{W}$

$P_{\text{in}} = P_{\text{out}}/\eta$

$I_L = \frac{P_{\text{in}}}{\sqrt{3}V_L \cos\theta}$

$V_p = V_L/\sqrt{3}$

$Z = V_p/I_p$

Step-by-Step Solution & Calculation:

1. Input Power:

$P_{\text{out}} = 12 \times 746 = 8,952\ \text{W}$

$P_{\text{in}} = 8,952/0.90 = 9,946.67\ \text{W}\ (9.95\ \text{kW})$

2. Line Current:

$$I_L = \frac{9,946.67}{\sqrt{3} \times 380 \times 0.85} = 17.78\ \text{A}$$

3. Phase Impedance (Y-connection: $V_p = 380/\sqrt{3} = 219.39\ \text{V}, I_p = I_L = 17.78\ \text{A}$):

- $|Z| = 219.39/17.78 = 12.34\ \Omega$
- $\theta = \arccos(0.85) = 31.79^\circ$
- $Z = 12.34\angle 31.79^\circ = 10.49 + j6.50\ \Omega$
- Resistance: $R = 10.49\ \Omega$
- Inductance: $X_L = 2\pi f L = 6.50 \implies L = 6.50/(2\pi \times 50) = 20.7\ \text{mH}.$

⚠️ Exam Pitfall / Pro Tip:Always convert output mechanical horsepower to electrical input power using efficiency ($P_{\text{in}} = P_{\text{out}}/\eta$) BEFORE calculating current!

CH06-08 Chapter 6 Level 2 (Past Exam / Calc) HW Ch.6 Problem 5 & Lecture Slides

Delta-Connected Motor Load Calculation

QUESTION / EXAM PROBLEM:

A 3ϕ, 10-hp motor is Δ-connected to a 3ϕ supply with $V_L = 380\ \text{V}, 50\ \text{Hz}$, $\eta = 85\%$, and $PF = 0.8$ lagging. Find: (1) Line current (I_L), (2) Phase current (I_p), (3) Per-phase Δ impedance (Z_Δ).

Key Formula / Operating Principle:
$P_{\text{in}} = (10 \times 746)/0.85 = 8,776.47\ \text{W}$
$I_L = \frac{P_{\text{in}}}{\sqrt{3} V_L \cos\theta}$
$I_p = I_L/\sqrt{3}, V_p = V_L = 380\ \text{V}$
$Z_\Delta = V_p/I_p$

Step-by-Step Solution & Calculation:

1. Input Power: $P_{\text{in}} = 7,460/0.85 = 8,776.47\ \text{W}$

2. Line Current:

$$I_L = \frac{8,776.47}{\sqrt{3} \times 380 \times 0.8} = 16.67\ \text{A}$$

3. Phase Quantities & Impedance:

- Phase current: $I_p = I_L/\sqrt{3} = 16.67/\sqrt{3} = 9.62\ \text{A}$
- Phase voltage: $V_p = V_L = 380\ \text{V}$
- Impedance magnitude: $|Z_\Delta| = 380/9.62 = 39.5\ \Omega$
- Angle: $\theta = \arccos(0.8) = 36.87^\circ$
- $Z_\Delta = 39.5\angle 36.87^\circ = 31.6 + j23.7\ \Omega$

($R_\Delta = 31.6\ \Omega, X_{L\Delta} = 23.7\ \Omega$).

⚠️ Exam Pitfall / Pro Tip:In Delta, $V_p = V_L$, but $I_p = I_L/\sqrt{3}$. Do not divide voltage by $\sqrt{3}$!

CH06-09 Chapter 6 Level 2 (Past Exam / Calc) 📌 Lecture Slides & Jele Formula Sheet

Two-Wattmeter Method for 3-Phase Power Measurement

QUESTION / EXAM PROBLEM:

Explain the Two-Wattmeter Method for measuring balanced 3-phase power. State the formulas for total active power (P), reactive power (Q), and power factor angle (θ) from meter readings W_1 and W_2.

Key Formula / Operating Principle:
$P_{\text{total}} = W_1 + W_2$
$Q_{\text{total}} = \sqrt{3}(W_1 - W_2)$
$\tan\theta = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$

Step-by-Step Solution & Calculation:

1. Total Real Power: $P = W_1 + W_2$

2. Total Reactive Power: $Q = \sqrt{3}(W_1 - W_2)$

3. Power Factor Angle: $\theta = \arctan\left(\sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}\right)$

4. Power Factor: $PF = \cos\theta$

Special Cases:
- If $W_1 = W_2$: $\theta = 0^\circ \implies PF = 1.0$ (purely resistive).
- If $W_2 = 0$: $PF = 0.5$ ($theta = 60^\circ$).
- If $W_2 < 0$: $PF < 0.5$ (one wattmeter reading turns negative).

⚠️ Exam Pitfall / Pro Tip:If $PF < 0.5$, one wattmeter deflects backward. In calculations, you MUST preserve the negative sign (e.g. $W_1 + (-W_2)$).

CH06-10 Chapter 6 Level 1 (Core Concept) 📌 HW Ch.6 Problem 1, 2, 3

Neutral Wire Current in Balanced vs Unbalanced Systems

QUESTION / EXAM PROBLEM:

What is the neutral current I_N in a balanced 4-wire Y-Y system? What causes current to flow in the neutral wire in an unbalanced system?

Key Formula / Operating Principle:
$I_N = -(I_a + I_b + I_c)$

Step-by-Step Solution & Calculation:

1. Balanced System: $I_a + I_b + I_c = 0$, therefore neutral current is strictly ZERO ($I_N = 0$). No current returns through the neutral line.

2. Unbalanced System: The phase currents do not sum to zero due to unequal load impedances ($Z_A \neq Z_B \neq Z_C$), so neutral current $I_N = -(I_a + I_b + I_c) \neq 0$ carries the return unbalance current.

⚠️ Exam Pitfall / Pro Tip:If the neutral wire is disconnected in an unbalanced load, the neutral point shifts ($V_{Nn} \neq 0$), causing severe overvoltage across the lighter load phase!

PART 2

Chapter 7: Electrical Machines (DC & AC)

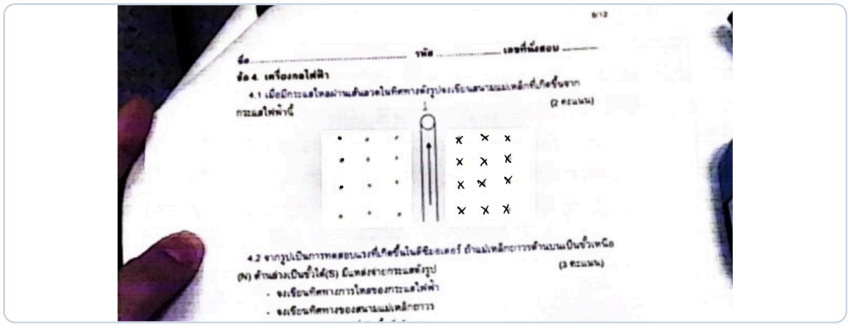
Right-Hand Rule, Lorentz Force, DC Shunt/Series Motors & Generators, Induction & Synchronous (15 Flashcards)

CH07-01 Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.1 (Page 28 - 2 Points)

Right-Hand Grip Rule for Magnetic Field around a Conductor

QUESTION / EXAM PROBLEM:

Current flows upward along a straight vertical wire. Using the right-hand grip rule, sketch/state the direction of the magnetic field on the left and right sides of the wire.



Exam Problem 4.1: Magnetic Field around Current Conductor

Key Formula / Operating Principle:
Right-Hand Grip Rule: Thumb = Current (I), Fingers = Field (B)

Step-by-Step Solution & Calculation:
Point right thumb upward along the current flow:
- Left side of the wire: Magnetic field lines point OUT OF THE PAGE (represented by dots •).
- Right side of the wire: Magnetic field lines point INTO THE PAGE (represented by crosses ×).

⚠️ Exam Pitfall / Pro Tip:Direct 2-point exam question! Remember: Dot (•) = coming towards you / out of paper; Cross (×) = going away from you / into paper.

CH07-02 Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.2 (Page 28 - 4 Points)

Lorentz Force on Current-Carrying Conductor in DC Motor

QUESTION / EXAM PROBLEM:

State the formula and rule used to determine the electromagnetic force developed on a conductor in a DC motor armature.

Key Formula / Operating Principle:
$\vec{F} = I(\vec{L} \times \vec{B}) \implies F = BIL\sin\alpha$

Step-by-Step Solution & Calculation:

1. Formula: $F = BIL\sin\alpha$, where B is flux density (Tesla), I is conductor current (A), L is active conductor length (m), and α is the angle between wire and magnetic field (usually 90°, so $F = BIL$).

2. Direction: Determined by Fleming's Left-Hand Rule for Motors:
- Thumb = Motion / Force (F)
- First finger = Magnetic Field (B, North to South)
- Second finger = Current (I, positive to negative).

⚠️ Exam Pitfall / Pro Tip:Use LEFT hand for motors (converting electrical to mechanical motion); use RIGHT hand for generators (inducing EMF).

CH07-03 Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.3 (Page 28 - 4 Points)

DC Machine Stator and Rotor Structural Components

QUESTION / EXAM PROBLEM:

What are the two main structural parts of a DC machine? Name at least 2 components found in each part.

Key Formula / Operating Principle:
Machine = Stator (Stationary) + Rotor (Rotating)

Step-by-Step Solution & Calculation:

1. Stator (Stationary Part):
- Frame / Yoke (cast iron outer housing and magnetic return path)
- Main Field Poles & Pole Shoes (laminated sheet steel)
- Field Windings (copper coils producing main flux Φ)
- End covers and brush holders.

2. Rotor / Armature (Rotating Part):
- Armature Core (laminated silicon steel with slots)
- Armature Windings (copper coils carrying load current)
- Commutator (segmented copper bars with mica insulation)
- Shaft, bearings, and cooling fan.

⚠️ Exam Pitfall / Pro Tip:Armature core is made of laminated silicon steel sheets specifically to minimize Eddy current losses and Hysteresis losses.

CH07-04 Chapter 7 Level 1 (Core Concept) 📌 Lecture Slides & Jele Formula Sheet

DC Machine Back EMF Formula & Physical Meaning

QUESTION / EXAM PROBLEM:

State the equation for induced armature back-EMF (E_a) in a DC machine and define all variables. How does it relate to motor speed?

Key Formula / Operating Principle:
$E_a = K_a \Phi \omega_m = \left(\frac{pZ}{2\pi a}\right) \Phi \omega_m$

Step-by-Step Solution & Calculation:
- E_a = Induced armature back EMF (V)
- $K_a = \frac{pZ}{2\pi a}$ = Machine design constant (p = poles, Z = total conductors, a = parallel paths)
- Φ = Magnetic flux per pole (Wb)
- ω_m = Mechanical angular velocity (rad/s) = $\frac{2\pi n}{60}$ (n in rpm).

Physical Meaning: In a motor, E_a opposes applied terminal voltage (V_t). As speed increases, E_a increases, limiting armature current $I_a = (V_t - E_a)/R_a$.

⚠️ Exam Pitfall / Pro Tip:At starting ($n = 0$), $E_a = 0$ V. Without starting resistors, inrush current $I_{start} = V_t/R_a$ can reach 10-20 times rated current!

CH07-05 Chapter 7 Level 1 (Core Concept) 📌 Lecture Slides & Homework 7

DC Machine Developed Torque & Converted Power

QUESTION / EXAM PROBLEM:

State the formula for developed electromagnetic torque (T_d) in a DC motor and the relationship between developed mechanical power and electrical power.

Key Formula / Operating Principle:

$T_d = K_a \Phi I_a$

$P_{\text{developed}} = E_a I_a = T_d \omega_m$

Step-by-Step Solution & Calculation:

1. Developed Torque: $T_d = K_a \Phi I_a$ **[$\mathrm{N \cdot m}$]. It is strictly proportional to field flux and armature current.**

2. Converted Power: $P_d = E_a I_a = T_d \omega_m$ **[W]. The internal electrical power converted equals mechanical power developed.**

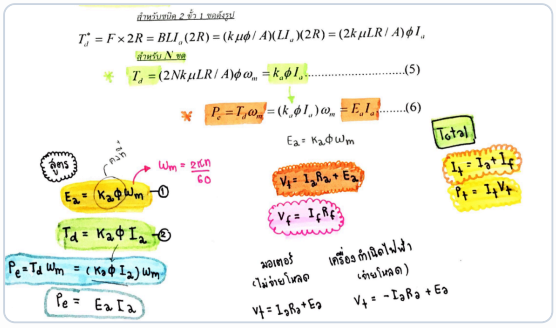
3. Shaft Output Power: $P_{\text{shaft}} = P_d - P_{\text{rotational losses}}$.

⚠️ Exam Pitfall / Pro Tip:Do not confuse developed torque (T_d) with shaft torque (T_{shaft}). Rotational losses (friction + windage) reduce output torque.

DC Shunt Motor Equivalent Circuit & KVL Equations

QUESTION / EXAM PROBLEM:

Draw the equivalent circuit of a DC Shunt Motor and state the KVL voltage loop equation and KCL current node equation.



DC Shunt Motor Equivalent Circuit & KVL

Key Formula / Operating Principle:

$V_t = E_a + I_a R_a$

$I_t = I_a + I_f$

$I_f = V_t / R_f$

Step-by-Step Solution & Calculation:

1. Field Circuit (Shunt across terminals):

$I_f = \frac{V_t}{R_f}$

2. Armature Circuit (KVL):

$$V_t = E_a + I_a R_a \implies E_a = V_t - I_a R_a$$

3. Current Relation (KCL at input terminal):

$$I_t = I_a + I_f \implies I_a = I_t - I_f$$

4. Converted Mechanical Power:

$$P_{\text{conv}} = E_a I_a$$

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⚠️ Exam Pitfall / Pro Tip:In a MOTOR, $V_t > E_a$ because the source drives current into the machine ($V_t = E_a + I_a R_a$).

CH07-07 Chapter 7 Level 2 (Past Exam / Calc) FunE-final.pdf Page 3 (Example 1)

DC Shunt Motor Solved Example (3 hp, 120 V)

QUESTION / EXAM PROBLEM:

A $3\ \text{hp}, 120\ \text{V}$ DC shunt motor has armature resistance $R_a = 0.2\ \Omega$, field resistance $R_f = 60\ \Omega$, and draws a full-load line current $I_t = 24.5\ \text{A}$. Calculate: (1) Field current I_f, (2) Armature current I_a, (3) Back EMF E_a, (4) Converted mechanical power P_{conv}, (5) Rotational loss if output is rated at $3\ \text{hp}$. ($1\ \text{hp} = 746\ \text{W}$)

Key Formula / Operating Principle:
$I_f = V_t / R_f$
$I_a = I_t - I_f$
$E_a = V_t - I_a R_a$
$P_{\text{conv}} = E_a I_a$
$P_{\text{rot}} = P_{\text{conv}} - P_{\text{out}}$

Step-by-Step Solution & Calculation:

1. Field Current:

$$I_f = \frac{120\ \text{V}}{60\ \Omega} = 2.0\ \text{A}$$

2. Armature Current:

$$I_a = I_t - I_f = 24.5 - 2.0 = 22.5\ \text{A}$$

3. Back EMF:

$$E_a = V_t - I_a R_a = 120 - (22.5 \times 0.2) = 120 - 4.5 = 115.5\ \text{V}$$

4. Converted Power:

$$P_{\text{conv}} = E_a I_a = 115.5\ \text{V} \times 22.5\ \text{A} = 2,598.75\ \text{W}$$

5. Rotational Loss:

$P_{\text{out}} = 3 \times 746 = 2,238\ \text{W}$

$$P_{\text{rot}} = P_{\text{conv}} - P_{\text{out}} = 2,598.75 - 2,238 = 360.75\ \text{W}$$

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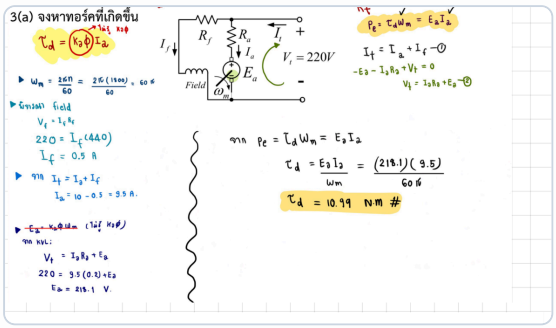
⚠️ Exam Pitfall / Pro Tip:Always separate internal converted power ($P_{conv} = E_a I_a$) from output shaft power (P_{out}). The difference is rotational friction and windage loss.

CH07-08 Chapter 7 🔴 Level 2 (Past Exam / Calc) 📌 FunE-final.pdf Page 5 (Example 3a & 3b)

DC Shunt Motor Speed Control & Series Resistance

QUESTION / EXAM PROBLEM:

A 220 V DC shunt motor has $R_a = 0.5\ \Omega$, $R_f = 110\ \Omega$. At full load, $I_t = 42$ A and speed $n = 1,200$ rpm. (1) Find developed torque T_d and constant $K_a\Phi$. (2) If series resistor $R_{\text{adj}} = 1.0\ \Omega$ is added in the armature branch and torque drops to $T_d' = 20\ \text{N}\cdot\text{m}$, calculate the new speed n'.



DC Shunt Motor Speed Control with Added Resistance

Key Formula / Operating Principle:

$I_f = V_t/R_f, I_a = I_t - I_f$

$E_a = V_t - I_aR_a$

$\omega_m = \frac{2\pi n}{60}, K_a\Phi = E_a/\omega_m$

$I_a' = T_d'/(K_a\Phi)$

$E_a' = V_t - I_a'(R_a + R_{\text{adj}})$

$n' = \frac{60E_a'}{2\pi(K_a\Phi)}$

Step-by-Step Solution & Calculation:

1. Initial Condition:

- $I_f = 220/110 = 2.0$ A
- $I_a = 42 - 2 = 40.0$ A
- $E_a = 220 - (40 \times 0.5) = 200.0$ V
- $\omega_m = \frac{2\pi(1200)}{60} = 125.66$ rad/s
- $K_a\Phi = \frac{E_a}{\omega_m} = \frac{200}{125.66} = 1.592\ \text{V}\cdot\text{s/rad}$
- $T_d = K_a\Phi I_a = 1.592 \times 40 = 63.66\ \text{N}\cdot\text{m}$

2. With $R_{\text{adj}} = 1.0\ \Omega$ and $T_d' = 20\ \text{N}\cdot\text{m}$:

- New armature current: $I_a' = \frac{20}{1.592} = 12.56$ A
- Total armature resistance: $R_{\text{tot}} = 0.5 + 1.0 = 1.5\ \Omega$
- New Back EMF: $E_a' = 220 - (12.56 \times 1.5) = 220 - 18.84 = 201.16$ V
- New Speed: $\omega_m' = \frac{201.16}{1.592} = 126.36$ rad/s
- In RPM: $n' = \frac{60 \times 126.36}{2\pi} = 1,206.6$ rpm.

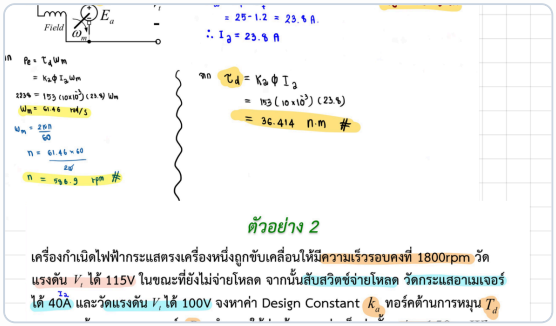
⚠️ Exam Pitfall / Pro Tip:When R_{adj} is inserted in the armature branch, total resistance becomes $R_a + R_{adj}$. Notice how reducing torque requirement allows speed to remain high even with added resistance.

CH07-09 Chapter 7 Level 2 (Past Exam / Calc) FunE-final.pdf Page 4 (Example 2)

DC Shunt Generator KVL, No-Load to Full-Load

QUESTION / EXAM PROBLEM:

A DC shunt generator driven at $1,800$ rpm produces terminal voltage $V_t = 115$ V at no load ($I_a = 0$). When loaded, it delivers $I_a = 40$ A at $V_t = 100$ V. Flux per pole is $\Phi = 150$ mWb. Find: (1) Machine constant K_a, (2) Armature resistance R_a, (3) Developed torque T_d.



DC Generator Circuit and KVL

Key Formula / Operating Principle:

No load: $E_a = V_t = 115$ V

$\omega_m = \frac{2\pi n}{60} = 188.5$ rad/s

$K_a = \frac{E_a}{\Phi \omega_m}$

Loaded: $E_a = V_t + I_a R_a \implies R_a = \frac{E_a - V_t}{I_a}$

$T_d = K_a \Phi I_a$

Step-by-Step Solution & Calculation:

1. Angular speed: $\omega_m = \frac{2\pi \times 1800}{60} = 188.5$ rad/s

2. At no load ($I_a = 0$), $E_a = V_t = 115$ V:

$$K_a = \frac{115}{150 \times 10^{-3} \times 188.5} = \frac{115}{28.27} = 4.07$$

3. Loaded operation: Assuming speed and flux remain constant, $E_a = 115$ V.

$$R_a = \frac{E_a - V_t}{I_a} = \frac{115 - 100}{40} = \frac{15}{40} = 0.375\ \Omega$$

4. Developed Torque:

$$T_d = K_a \Phi I_a = 4.07 \times (150 \times 10^{-3}) \times 40 = 24.42\ \text{N} \cdot \text{m}$$

.

⚠️ Exam Pitfall / Pro Tip:Crucial distinction: In a GENERATOR: $E_a = V_t + I_a R_a$ ($E_a > V_t$). In a MOTOR: $E_a = V_t - I_a R_a$ ($V_t > E_a$).

CH07-10 Chapter 7 🔴 Level 2 (Past Exam / Calc) 📌 HW Ch.7 Problem 4 & Talui Homework Page 1

DC Shunt Motor Equivalent Circuit & Speed/Torque Calculation

QUESTION / EXAM PROBLEM:

A 2-pole, 2.5 hp, 100 V DC shunt motor has $K_a = 153$, full-load total current $I_t = 25.0\ \text{A}$, armature resistance $R_a = 0.75\ \Omega$, field flux $\Phi = 10.0\ \text{mWb}$, and field resistance $R_f = 440\ \Omega$. Calculate: (1) Armature current (I_a), (2) Back EMF (E_a), (3) Motor speed (n in rpm), (4) Developed torque (T_d).

Key Formula / Operating Principle:

$I_f = V_t / R_f$
$I_a = I_t - I_f$
$E_a = V_t - I_a R_a$
$\omega_m = E_a / (K_a \Phi) \implies n = \frac{60\omega_m}{2\pi}$
$T_d = K_a \Phi I_a$

Step-by-Step Solution & Calculation:

1. Field current: $I_f = 100/440 = 0.23\ \text{A}$

2. Armature current: $I_a = I_t - I_f = 25.0 - 0.23 = 24.77\ \text{A}$

3. Back EMF: $E_a = V_t - I_a R_a = 100 - (24.77 \times 0.75) = 100 - 18.58 = 81.42\ \text{V}$

4. Speed calculation:

$$E_a = K_a \Phi \omega_m = 153 \times (10 \times 10^{-3}) \times \left(\frac{2\pi n}{60}\right)$$

$$81.42 = 1.53 \times 0.10472 \times n = 0.16022 \times n \implies n = 508.17\ \text{rpm}$$

5. Developed Torque:

$$T_d = K_a \Phi I_a = 153 \times (10 \times 10^{-3}) \times 24.77 = 37.9\ \text{N} \cdot \text{m}$$

.

⚠️ Exam Pitfall / Pro Tip:Always subtract field current I_f from total supply current I_t before computing $I_a R_a$ drop! Field current does not flow through armature.

CH07-11 Chapter 7 Level 2 (Past Exam / Calc) 📌 HW Ch.7 Problem 5 & Talui Homework Page 2-3

DC Series Motor Comparison & Runaway Hazard

QUESTION / EXAM PROBLEM:

A 2.5 hp series motor has $K_a = 153, R_a = 0.75\ \Omega, \Phi = 10.0\ \text{mWb}, R_f = 440\ \Omega$. Find the required terminal voltages: (a) V_{t1} to give the same speed ($508.17\ \text{rpm}$) as Problem 4, (b) V_{t2} to give the same torque ($37.9\ \text{N} \cdot \text{m}$) as Problem 4.

Key Formula / Operating Principle:
$V_t = I_a(R_a + R_f) + E_a$
(a) Same speed: $E_a = 81.42\ \text{V}, I_a = P/E_a = 1865/81.42 = 22.91\ \text{A}$
(b) Same torque: $I_a = 24.77\ \text{A}, E_a = P/I_a = 1865/24.77 = 75.3\ \text{V}$

Step-by-Step Solution & Calculation:

(a) For same speed ($n = 508.17\ \text{rpm}$):

$E_a = 81.42\ \text{V}$

$I_a = P_e/E_a = 1865/81.42 = 22.91\ \text{A}$

$$V_{t1} = I_a(R_f + R_a) + E_a = 22.91(440 + 0.75) + 81.42 = 10,181\ \text{V} = 10.18\ \text{kV}$$

(b) For same torque ($T_d = 37.9\ \text{N} \cdot \text{m}$):

$I_a = 24.77\ \text{A}$

$E_a = P_e/I_a = 1865/24.77 = 75.3\ \text{V}$

$$V_{t2} = I_a(R_f + R_a) + E_a = 24.77(440.75) + 75.3 = 10,993\ \text{V} \approx 11.0\ \text{kV}$$

(c) Can same speed and same torque happen simultaneously? NO, because speed and torque uniquely dictate E_a and I_a.

⚠️ Exam Pitfall / Pro Tip:Why must a DC series motor never run at no load? Because $I_a \to 0 \implies \Phi \to 0 \implies \omega_m \approx V_t/(K_a\Phi) \to \infty$ (motor dangerously overspeeds and tears itself apart).

CH07-12 Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.4 (Page 27 - 8 Points)

Squirrel Cage Induction Motor & Reversing Rotation

QUESTION / EXAM PROBLEM:

What is a squirrel-cage rotor in a 3-phase induction motor? Explain its operating principle and describe how to reverse its direction of rotation.

Key Formula / Operating Principle:
Swap any 2 supply lines (Phase A and Phase B)

Step-by-Step Solution & Calculation:

1. Structure: Rotor composed of thick copper or aluminum conductor bars embedded in a laminated steel core, permanently short-circuited at both ends by heavy conductive end-rings (resembling a squirrel exercise wheel).

2. Operating Principle: 3-phase AC in stator windings produces a Rotating Magnetic Field (RMF). The RMF sweeps across the rotor bars, inducing an EMF and heavy circulating currents. Interaction between rotor currents and the RMF produces driving torque following Lenz's Law.

3. Reversing Direction: Swap ANY TWO of the three incoming supply lines (e.g. swap A and B). This reverses the phase sequence from A-B-C to A-C-B, reversing the RMF rotation direction.

⚠️ Exam Pitfall / Pro Tip:Direct 8-point exam question! Never say 'reverse DC polarity' because this is an AC motor without brushes or commutators.

CH07-13 Chapter 7 Level 1 (Core Concept) 📌 Lecture Slides & Jele Formula Sheet

Synchronous Speed, Rotor Speed, and Slip

QUESTION / EXAM PROBLEM:

Define Synchronous Speed (N_s) and Slip (s) in a 3-phase induction motor. Why can an induction motor never run at synchronous speed under normal motoring conditions?

Key Formula / Operating Principle:
$N_s = \frac{120f}{p} \quad [\text{rpm}]$
$s = \frac{N_s - N_r}{N_s} = \frac{\omega_s - \omega_r}{\omega_s}$

Step-by-Step Solution & Calculation:

1. Synchronous Speed: $N_s = \frac{120f}{p}$ is the rotational speed of the stator magnetic field (f = grid frequency in Hz, p = total poles).

2. Slip: $s = \frac{N_s - N_r}{N_s}$, typically $2\% - 5\%$ at rated full load.

3. Why $N_r < N_s$: If the rotor ran at synchronous speed ($N_r = N_s$), the relative motion between rotor bars and magnetic field would become zero ($d\Phi/dt = 0$). No EMF would be induced, zero rotor current would flow, and developed torque would drop to zero. The rotor must slip behind N_s to produce torque.

⚠️ Exam Pitfall / Pro Tip:If p is given as pole pairs P, the formula is $N_s = \frac{60f}{P}$. Always check whether the question specifies poles or pole pairs.

CH07-14 Chapter 7 Level 1 (Core Concept) 📌 AC Machines Slides P.12-14

Types of Single-Phase Induction Motors

QUESTION / EXAM PROBLEM:

Single-phase induction motors are not self-starting. Name the 3 common types used to create starting torque and describe their operating features.

Key Formula / Operating Principle:
Shaded-Pole | Capacitor-Start | Universal Motor

Step-by-Step Solution & Calculation:

1. Shaded-Pole Motor:
- Uses copper shading rings on pole faces to create a delayed phase shift in magnetic flux.
- Low starting torque, simple, low cost; used in small desk fans and blowers.

2. Capacitor-Start Motor:
- Uses an auxiliary starting winding in series with a capacitor and centrifugal switch.
- High starting torque ($200 - 350\%$ of rated); switch disconnects capacitor at $\approx 75\%$ speed; used in air compressors, water pumps, refrigerators.

3. Universal Motor:
- Series-wound motor that operates on both AC and DC.
- Very high speed (up to 20,000 rpm) and high torque-to-weight ratio; used in vacuum cleaners and power drills.

⚠️ Exam Pitfall / Pro Tip:Pure single-phase AC creates a pulsating (standing) magnetic field, not a rotating magnetic field. It requires an auxiliary phase or shading ring to start rotating.

CH07-15 Chapter 7 Level 1 (Core Concept) 📌 AC Machines Slides P.15 & HW7 Problem 3

Synchronous Motor vs Induction Motor Operation

QUESTION / EXAM PROBLEM:

Compare the rotor construction and operating speed of a Synchronous Motor with an Induction Motor. How does a synchronous motor maintain exact synchronous speed?

Key Formula / Operating Principle:
$N_r = N_s$ (Synchronous) vs $N_r < N_s$ (Induction)

Step-by-Step Solution & Calculation:

1. Synchronous Motor:

- Rotor is excited by an external DC source (or permanent magnets) to establish fixed magnetic poles.
- Rotor magnetic poles magnetically lock with the stator's Rotating Magnetic Field (RMF).
- Runs at exactly synchronous speed ($N_r = N_s = \frac{120f}{p}$) under all steady-load conditions (zero slip, $s = 0$).

2. Induction Motor:

- Rotor currents are induced purely by relative motion between RMF and rotor conductors.
- Must always run below synchronous speed ($N_r < N_s$) to produce torque.

⚠️ Exam Pitfall / Pro Tip:Synchronous motors can also operate at leading power factor by over-exciting the DC field, functioning as a Synchronous Condenser for power factor correction.

PART 3

Chapter 8: Transformers

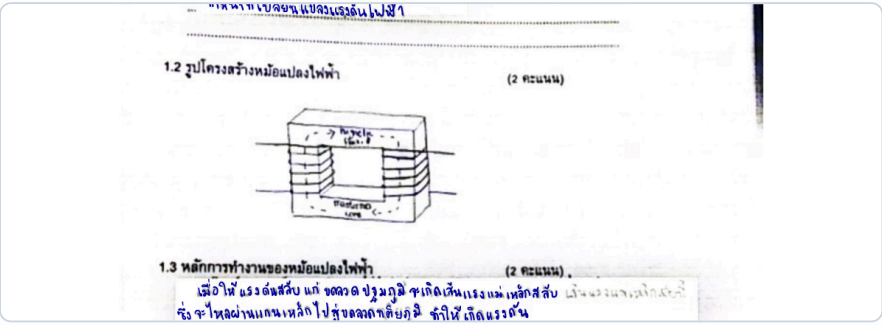
Mutual Induction, Induced EMF Formula, 25 kVA Exam Problem, Audio Matching, Efficiency (11 Flashcards)

CH08-01 Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.1 & 1.2 (Page 32 - 4 Points)

Transformer Definition, Purpose & Structure

QUESTION / EXAM PROBLEM:

What is an electrical transformer? What is its primary function and structure? Draw/describe its essential construction.



Exam Problem 1.2: Transformer Magnetic Core & Coils

Key Formula / Operating Principle:
Static machine transferring AC electrical energy via mutual induction.

Step-by-Step Solution & Calculation:

1. Definition & Purpose: A static electrical machine that transfers AC electrical power between circuits at the same frequency, stepping voltage up or down to minimize transmission loss or match load requirements.

2. Structure: Consists of:

- High-permeability laminated silicon steel core (provides closed magnetic circuit)
- Primary winding (N_1 turns) connected to input AC source
- Secondary winding (N_2 turns) connected to output load.

⚠️ Exam Pitfall / Pro Tip:A transformer changes voltage and current levels, but NEVER changes power frequency ($f_1 = f_2 = 50\ \text{Hz}$).

CH08-02 Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.3 (Page 32 - 4 Points)

Working Principle & Induced EMF Equation

QUESTION / EXAM PROBLEM:

Explain the working principle of a transformer and write Faraday's induced EMF formula including the derivation constant.

Key Formula / Operating Principle:
$E_{\text{rms}} = 4.44 f N \Phi_m = \sqrt{2}\pi f N \Phi_m$

Step-by-Step Solution & Calculation:

1. Working Principle: Mutual electromagnetic induction according to Faraday's law. When AC voltage V_1 is applied to the primary winding, it drives alternating current that establishes a time-varying magnetic flux $\Phi(t) = \Phi_m \sin(\omega t)$ in the core. This flux links both windings, inducing EMF $e_1 = -N_1 \frac{d\Phi}{dt}$ and $e_2 = -N_2 \frac{d\Phi}{dt}$.

2. RMS EMF Equation: Since $\frac{d}{dt}[\sin(\omega t)] = \omega \cos(\omega t)$, peak value is $E_m = 2\pi f N \Phi_m$. Dividing by $\sqrt{2}$ yields:

$$E_{\text{rms}} = \frac{2\pi}{\sqrt{2}} f N \Phi_m = \sqrt{2}\pi f N \Phi_m = 4.44 f N \Phi_m$$

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⚠️ Exam Pitfall / Pro Tip:Remember the constant 4.44 comes from $\sqrt{2}\pi \approx 1.4142 \times 3.1416 = 4.4428$.

CH08-03 Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.4 (Page 32 - 2 Points)

Ideal Transformer 4 Essential Properties

QUESTION / EXAM PROBLEM:

List the four fundamental assumptions/properties of an Ideal Transformer.

Key Formula / Operating Principle:
1. Zero resistance, 2. Zero leakage, 3. Zero core loss, 4. Infinite permeability

Step-by-Step Solution & Calculation:

1. Zero winding resistance: $R_1 = 0, R_2 = 0$ (no I^2R copper losses; $V_1 = E_1, V_2 = E_2$).

2. Zero leakage flux: All magnetic flux is completely confined within the core and links 100% of both primary and secondary windings ($L_1 = 0, L_2 = 0$).

3. Zero core loss: No hysteresis loss and no eddy current loss in the core ($P_{\text{core}} = 0$, efficiency $\eta = 100\%$).

4. Infinite core permeability: $\mu_r \to \infty$, meaning zero magnetizing current is required to establish core flux.

⚠️ Exam Pitfall / Pro Tip:Direct 2-point exam question! Make sure to list all 4 properties clearly.

CH08-04 Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.5 (Page 32 - 4 Points)

Power Losses in a Practical Transformer

QUESTION / EXAM PROBLEM:

What types of power losses occur in a real transformer? Name the 2 primary loss categories and where they occur.

Key Formula / Operating Principle:
$P_{\text{loss}} = P_{\text{copper}} + P_{\text{core}} = (I_1^2 R_1 + I_2^2 R_2) + (P_h + P_e)$

Step-by-Step Solution & Calculation:

1. Copper Losses (Winding Losses): Occur in the copper windings due to Joule heating: $P_{\text{cu}} = I_1^2 R_1 + I_2^2 R_2$. Variable loss proportional to $(I_{\text{load}})^2$.

2. Core / Iron Losses: Occur in the magnetic core, nearly constant at all load levels:

- Hysteresis Loss: Energy lost in continuously reversing magnetic domains ($P_h \propto f B_m^{1.6}$).
- Eddy Current Loss: Circulating currents induced in the conductive iron core ($P_e \propto f^2 B_m^2 t^2$). Minimized by using thin laminated sheets insulated by varnish.

⚠️ Exam Pitfall / Pro Tip:Core loss depends on voltage and frequency (constant 24/7); copper loss depends on load current squared.

CH08-05 Chapter 8 Level 2 (Past Exam / Calc) Final Exam Problem 1.6 (Page 31 - 12 Points)

Step-Down Transformer Parameter Calculation

QUESTION / EXAM PROBLEM:

A single-phase transformer rated at $25\ \text{kVA}, 50\ \text{Hz}$ has primary winding $N_1 = 500$ turns and secondary winding $N_2 = 50$ turns. The primary winding is connected to $V_1 = 3,000\ \text{V}$. Calculate: (a) Turns ratio (a), (b) Full-load rated primary and secondary currents (I_1, I_2), (c) Induced secondary voltage (V_2), (d) Maximum magnetic flux in core (Φ_m).

Key Formula / Operating Principle:
$a = N_1/N_2$
$I_1 = S/V_1, I_2 = S/V_2$
$V_2 = V_1/a$
$\Phi_m = \frac{V_1}{4.44 f N_1}$

Step-by-Step Solution & Calculation:

(a) Turns Ratio:

$$a = \frac{N_1}{N_2} = \frac{500}{50} = 10$$

(b) Rated Full-Load Currents:

$$I_1 = \frac{S_{\text{rated}}}{V_1} = \frac{25,000}{3,000} = 8.33\ \text{A}$$

$$I_2 = a \times I_1 = 10 \times 8.33 = 83.33\ \text{A}$$

(c) Secondary Voltage:

$$V_2 = \frac{V_1}{a} = \frac{3,000}{10} = 300\ \text{V}$$

(d) Maximum Magnetic Flux:

$$\Phi_m = \frac{V_1}{4.44 f N_1} = \frac{3,000}{4.44 \times 50 \times 500} = \frac{3,000}{111,000} = 0.0270\ \text{Wb} = 27.0\ \text{mWb}$$

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⚠️ Exam Pitfall / Pro Tip:Exact problem from past exam page 31 (12 points total)! Write all steps and units clearly (A, V, Wb).

CH08-06 Chapter 8 Level 2 (Past Exam / Calc) HW Ch.8 Problem 2 & Talui Homework Page 4

140V/80V Transformer Turns & Currents

QUESTION / EXAM PROBLEM:

A single-phase transformer has primary and secondary voltages $V_1 = 140\ \text{V}$ and $V_2 = 80\ \text{V}$. Primary current is $I_1 = 8\ \text{A}$. The secondary winding has $N_2 = 800\ \text{turns}$. Find: (1) Primary winding turns N_1, (2) Secondary current I_2, (3) Turns ratio a.

Key Formula / Operating Principle:
$a = V_1/V_2 = N_1/N_2$
$I_2 = a \times I_1$

Step-by-Step Solution & Calculation:

1. Turns Ratio:

$$a = \frac{V_1}{V_2} = \frac{140}{80} = 1.75$$

2. Primary Winding Turns:

$$N_1 = a \times N_2 = 1.75 \times 800 = 1,400\ \text{turns}$$

3. Secondary Winding Current:

$$I_2 = a \times I_1 = 1.75 \times 8\ \text{A} = 14.0\ \text{A}$$

.

⚠️ Exam Pitfall / Pro Tip:Notice that voltage drops by 1.75 times, so current increases by 1.75 times to conserve total volt-amperes ($S = V_1 I_1 = V_2 I_2 = 1,120\ \text{VA}$).

CH08-07 Chapter 8 Level 2 (Past Exam / Calc) HW Ch.8 Problem 3 & Talui Homework Page 4

200 kVA, 2200V/220V Transformer Full Calculations

QUESTION / EXAM PROBLEM:

A single-phase transformer rated at $200\ \text{kVA}, 2,200\ \text{V}/220\ \text{V}, 50\ \text{Hz}$ has primary winding $N_1 = 200\ \text{turns}$. Find: (1) Rated primary and secondary currents I_1, I_2, (2) Secondary winding turns N_2, (3) Maximum magnetic flux in core Φ_m.

Key Formula / Operating Principle:
$I_1 = S/V_1, I_2 = S/V_2$
$a = V_1/V_2 = N_1/N_2$
$\Phi_m = \frac{V_1}{4.44 f N_1}$

Step-by-Step Solution & Calculation:

1. Rated Currents:

$$I_1 = \frac{200,000\ \text{VA}}{2,200\ \text{V}} = 90.91\ \text{A}$$

$$I_2 = \frac{200,000\ \text{VA}}{220\ \text{V}} = 909.09\ \text{A}$$

2. Secondary Turns:

Turns ratio $a = \frac{2,200}{220} = 10$

$$N_2 = \frac{N_1}{a} = \frac{200}{10} = 20\ \text{turns}$$

3. Maximum Core Flux:

$$\Phi_m = \frac{V_1}{4.44 f N_1} = \frac{2,200}{4.44 \times 50 \times 200} = \frac{2,200}{44,400} = 0.0495\ \text{Wb} = 49.5\ \text{mWb}$$

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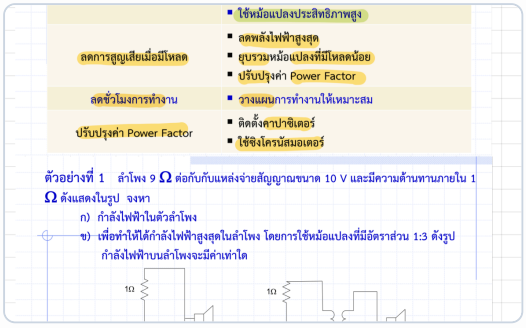
⚠️ Exam Pitfall / Pro Tip:Always use rated apparent power in Volt-Amperes ($200\ \text{kVA} = 200,000\ \text{VA}$), not kilo-units.

CH08-08 Chapter 8 🔴 Level 2 (Past Exam / Calc) 📌 Ch07Transformer.pdf Example 1 (Page 18) & FunE P.7

Impedance Matching with Transformer for Maximum Power

QUESTION / EXAM PROBLEM:

A 10 V audio source with internal source resistance $R_s = 1\ \Omega$ is connected to a speaker load $R_L = 9\ \Omega$. (1) Find load power without transformer. (2) If an ideal transformer with turns ratio $a = 1/3$ is inserted between source and speaker, find the reflected load resistance and the new load power.



Audio Source Impedance Matching via 1:3 Transformer

Key Formula / Operating Principle:

$R'_L = a^2 R_L = \left(\frac{N_1}{N_2}\right)^2 R_L$

$P = I^2 R$

Step-by-Step Solution & Calculation:

1. Without Transformer:

$$I = \frac{V_s}{R_s + R_L} = \frac{10}{1+9} = 1.0\ \text{A}$$

$$P_{\text{speaker}} = I^2 R_L = (1.0)^2 \times 9 = 9.0\ \text{W}$$

2. With Transformer ($a = 1/3$):

Reflected load resistance seen by source:

$$R'_L = a^2 R_L = \left(\frac{1}{3}\right)^2 \times 9 = \frac{1}{9} \times 9 = 1.0\ \Omega$$

Notice $R'_L = R_s = 1.0\ \Omega$ (perfect Maximum Power Transfer match!).

New primary current:

$$I_1 = \frac{V_s}{R_s + R'_L} = \frac{10}{1+1} = 5.0\ \text{A}$$

Power delivered to reflected load (and thus speaker, since transformer is ideal):

$$P_{\text{speaker}} = I_1^2 R'_L = (5.0)^2 \times 1.0 = 25.0\ \text{W}$$

Power increased dramatically from 9 W to 25 W!

⚠️ Exam Pitfall / Pro Tip:Reflected impedance scales as a^2: $Z_1 = a^2 Z_2$. In audio systems, transformers match unequal impedances to maximize transferred power.

CH08-09 Chapter 8 Level 2 (Past Exam / Calc) HW Ch.8 Problem 4 & Talui Homework Page 5

DC Voltage Connection to Transformer Primary Hazard

QUESTION / EXAM PROBLEM:

What happens if a DC voltage (e.g. 200 V DC) is connected to the primary winding of a transformer? What will the secondary voltage and current be?

Key Formula / Operating Principle:
$v_2 = -N_2 \frac{d\Phi}{dt} = 0$, $I_1 = \frac{V_{dc}}{R_1}$ (catastrophic overcurrent)

Step-by-Step Solution & Calculation:

1. Secondary Voltage: $V_2 = 0$ V. Because DC voltage creates static, time-invariant magnetic flux ($\frac{d\Phi}{dt} = 0$), no voltage is induced in the secondary winding.

2. Primary Hazard: On DC, inductive reactance is zero ($X_L = 2\pi f L = 0$ since $f = 0$). There is no counter-EMF to oppose the source. Primary current is limited solely by tiny copper wire resistance R_1: $I_1 = V_{dc}/R_1 \gg I_{rated}$. The primary winding rapidly overheats and burns out.

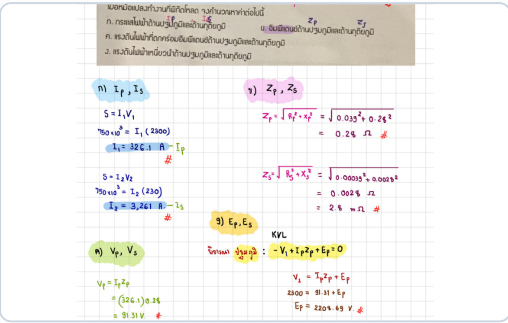
⚠️ Exam Pitfall / Pro Tip:Exam favorite concept! Always state: 1) Secondary output is zero, and 2) Primary winding burns out.

CH08-10 Chapter 8 🔴 Level 2 (Past Exam / Calc) 📌 FunE-final.pdf Page 10 (Example 5)

Practical Transformer Equivalent Circuit & Efficiency

QUESTION / EXAM PROBLEM:

A $10\ \text{kVA}, 2200/220\ \text{V}, 50\ \text{Hz}$ practical transformer has primary parameters $R_1 = 3.5\ \Omega, X_1 = 6.2\ \Omega$ and secondary parameters $R_2 = 0.035\ \Omega, X_2 = 0.062\ \Omega$. Core loss is $P_{\text{core}} = 150\ \text{W}$. Find equivalent impedance referred to primary, full-load copper loss, and efficiency at $PF = 1.0$ and $PF = 0.8$ lagging.



Practical Transformer Equivalent Circuit & KVL

Key Formula / Operating Principle:

$a = V_1/V_2 = 10$

$R_{eq1} = R_1 + a^2 R_2, X_{eq1} = X_1 + a^2 X_2$

$I_1 = S/V_1$

$P_{\text{cu}} = I_1^2 R_{eq1}$

$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{cu}} + P_{\text{core}}} \times 100\%$

Step-by-Step Solution & Calculation:

1. Equivalent Parameters referred to Primary:

$a = 2200/220 = 10$

- $R_{eq1} = 3.5 + (10)^2(0.035) = 3.5 + 3.5 = 7.0\ \Omega$
- $X_{eq1} = 6.2 + (10)^2(0.062) = 6.2 + 6.2 = 12.4\ \Omega$
- $Z_{eq1} = 7.0 + j12.4\ \Omega$

2. Full-Load Primary Current & Copper Loss:

$I_1 = 10,000/2,200 = 4.545\ \text{A}$

$$P_{\text{cu}} = (4.545)^2 \times 7.0 = 144.6\ \text{W}$$

3. Efficiency at Unity PF ($PF = 1.0$):

$P_{\text{out}} = 10,000 \times 1.0 = 10,000\ \text{W}$

$$\eta = \frac{10,000}{10,000 + 144.6 + 150} \times 100\% = \frac{10,000}{10,294.6} \times 100\% = 97.14\%$$

4. Efficiency at $PF = 0.8$ lagging:

$P_{\text{out}} = 10,000 \times 0.8 = 8,000\ \text{W}$

$$\eta = \frac{8,000}{8,000 + 144.6 + 150} \times 100\% = \frac{8,000}{8,294.6} \times 100\% = 96.45\%$$

.

⚠️ Exam Pitfall / Pro Tip:Notice how lower power factor degrades transformer efficiency because the real power output drops while I^2R copper loss remains at full rating.

CH08-11 Chapter 8 Level 1 (Core Concept) 📌 FunE-final.pdf Page 11 & Lecture Slides

Energy Conservation in Transformer Operations

QUESTION / EXAM PROBLEM:

State three practical methods to save electrical energy in industrial transformer substations.

Key Formula / Operating Principle:
1. Reduce no-load loss | 2. Consolidate light loads | 3. Power Factor correction

Step-by-Step Solution & Calculation:

1. De-energize Unloaded Transformers: Core losses occur 24 hours a day regardless of load. Switching off backup or redundant transformers during weekends/nights eliminates standing core loss.

2. Load Consolidation: Operating two transformers at 20% load is far less efficient than shutting one down and running a single unit at 40-50% load near its peak efficiency point.

3. Power Factor Correction: Installing shunt capacitor banks on the secondary side cancels inductive VARs, reducing secondary line current, which cuts I^2R winding copper losses and lowers transformer operating temperature.

⚠️ Exam Pitfall / Pro Tip:Peak efficiency occurs when copper loss equals core loss ($P_{cu} = P_{core}$), which is typically engineered around 50% to 75% rated load.

PART 4

Chapter 9: Power Transmission Systems & Energy Conservation

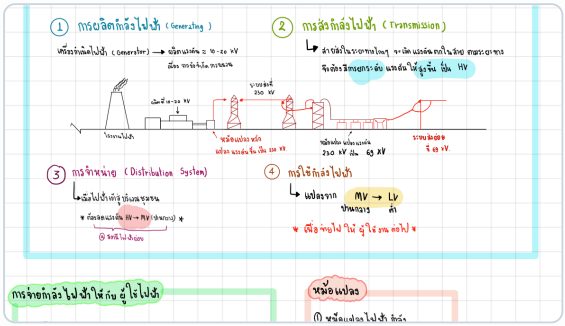
Thai Grid Voltage Stages, EGAT/MEA/PEA, Power Plants, Tower Classifications, Energy Conservation (10 Flashcards)

CH09-01 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.3 (Page 25 - 4 Points)

Structure of Electrical Power System & Voltage Levels

QUESTION / EXAM PROBLEM:

State the four stages of a power system and classify these standard Thai grid voltages: 500 kV, 22 kV, 380 V, and residential household voltage.



4 Stages of Electric Power Transmission System

Key Formula / Operating Principle:
Generation -> Transmission -> Distribution -> Utilization

Step-by-Step Solution & Calculation:

1. Generation: 10 - 20 kV (alternator terminals)

2. Transmission: 115 kV, 230 kV, 500 kV

3. Distribution: 22 kV, 24 kV, 33 kV

4. Utilization: 380/400 V (3ϕ) and 220/230 V (1ϕ)

Classifications for Exam Problem 5.3:

- 500 kV: Extra High Voltage (EHV)
- 22 kV: Medium Voltage (MV)
- 380 V: Low / Utilization Voltage
- Residential Household Voltage: 220 V (or modern 230 V).

⚠️ Exam Pitfall / Pro Tip:Why transmit at Extra High Voltage (500 kV)? Because $P = \sqrt{3}V_L I_L \cos\theta$. Stepping up V reduces line current I proportionally, minimizing I^2R transmission wire loss by $(1/a)^2$.

CH09-02 Chapter 9 Level 1 (Core Concept) 📌 Lecture Slides & Jele Formula Sheet

Three Electricity Authorities in Thailand & Roles

QUESTION / EXAM PROBLEM:

Name the three state electricity authorities in Thailand and state their specific geographic responsibilities.

Key Formula / Operating Principle:
EGAT (Generation/Transmission) | MEA (BKK+2) | PEA (74 Provinces)

Step-by-Step Solution & Calculation:

1. EGAT (Electricity Generating Authority of Thailand / กฟผ.): Responsible for power generation and nationwide high-voltage transmission grid (115-500 kV); sells wholesale power to MEA, PEA, and industrial direct clients.

2. MEA (Metropolitan Electricity Authority / กฟน.): Distributes electricity in 3 central provinces: Bangkok, Nonthaburi, and Samut Prakan.

3. PEA (Provincial Electricity Authority / กฟภ.): Distributes electricity across the remaining 74 provinces in Thailand.

⚠️ Exam Pitfall / Pro Tip:Guaranteed exam question! Remember MEA covers strictly Bangkok, Nonthaburi, and Samut Prakan.

CH09-03 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.1 (Page 25 - 2 Points)

Fuel Mix for Power Generation in Thailand

QUESTION / EXAM PROBLEM:

What is the most widely used fuel source for electricity generation in Thailand, and what fuel source is used least?

Key Formula / Operating Principle:
Most: Natural Gas (~55-60%) | Least: Fuel Oil / Heavy Fuel Oil

Step-by-Step Solution & Calculation:

- Most used: Natural Gas (ก๊าซธรรมชาติ) accounts for over 55-60% of total electrical energy in Thailand (Gulf of Thailand, Myanmar, and imported LNG).
- Followed by: Coal / Lignite (Mae Moh power plant) and Renewable/Hydro imports from Lao PDR.
- Least used: Heavy fuel oil (น้ำมันเตา) and diesel, due to extremely high operating costs and environmental impact.

⚠️ Exam Pitfall / Pro Tip:Do not answer solar or oil as the main source; it is overwhelmingly Natural Gas.

CH09-04 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.2 (Page 25 - 4 Points)

Base Load vs Peak Load Power Plants & Startup Times

QUESTION / EXAM PROBLEM:

Compare Base Load and Peak Load power plants. Which plant type has the shortest startup time, and which has the longest?

Key Formula / Operating Principle:
Base Load: Steam/Coal (6-8 hrs) | Peak Load: Hydro (<5 mins)

Step-by-Step Solution & Calculation:

1. Base Load Plants: Run continuously 24/7 to supply minimum baseline grid demand. High capital cost, low running cost.

- Longest Startup Time: Steam thermal power plants (e.g. Mae Moh coal/lignite) take 6 - 8 hours to bring boilers up to operating pressure from cold start.

2. Peak Load Plants: Dispatched quickly to handle rapid spikes in daily demand.

- Shortest Startup Time: Hydroelectric power plants (**โรงไฟฟ้าพลังน้ำ**) can synchronize and deliver full load in less than 5 minutes. Gas turbines take 10 - 15 minutes.

⚠️ Exam Pitfall / Pro Tip:Direct 4-part exam question: Shortest = Hydro; Longest = Steam; Base = Steam/Coal; Peak = Hydro/Gas turbine.

CH09-05 Chapter 9 Level 1 (Core Concept) 📌 FunE-final.pdf Page 14 & Lecture Slides

Transmission Tower Classifications & Structures

QUESTION / EXAM PROBLEM:

Describe the structural features and conductor configurations used on Thai grid transmission towers at 500 kV, 230 kV, and 115 kV.



Transmission Tower Classifications (500 kV, 230 kV, 115 kV, 69 kV)

Key Formula / Operating Principle:
500 kV (Quadruple bundle) | 230 kV (Twin bundle) | 115 kV (Single conductor)

Step-by-Step Solution & Calculation:

1. 500 kV Towers: Massive steel lattice towers. Use Quadruple Bundle Conductors (4 sub-conductors spaced per phase) to expand effective conductor diameter, suppressing Corona discharge and Audible noise, with ~28-35 insulator discs per string.

2. 230 kV Towers: Double-circuit lattice towers using Twin Bundle Conductors (2 sub-conductors per phase) with ~14-17 insulator discs.

3. 115 kV Towers: Single-conductor lattice or spun concrete poles with ~7-9 insulator discs, linking sub-transmission substations to city borders.

⚠️ Exam Pitfall / Pro Tip:Bundle conductors are used at 500 kV and 230 kV primarily to reduce high-voltage Corona discharge loss and radio interference.

CH09-06 Chapter 9 Level 1 (Core Concept) 📌 FunE-final.pdf Page 13 & PEA/MEA Regulations

Customer Categories & Supply Voltage Thresholds

QUESTION / EXAM PROBLEM:

How are electricity consumers categorized by supply voltage in Thailand? State the voltage levels supplied to: (1) Residential households, (2) Medium business (30-999 kVA), (3) Large industrial (>10,000 kVA).

Key Formula / Operating Principle:
<30 kVA: LV 230/400 V | 30-999 kVA: MV 22/33 kV | >10 MVA: HV 115 kV

Step-by-Step Solution & Calculation:

1. Low Voltage Customers (< 30 kVA):
- Residential households and small shops.
- Supplied at $230\ \text{V}$ (1ϕ, 2-wire) or $400/230\ \text{V}$ (3ϕ, 4-wire).

2. Medium Voltage Customers (30 - 999 kVA):
- Commercial buildings, hotels, small factories.
- Supplied at $22\ \text{kV}$ or $33\ \text{kV}$ (PEA) / $12\ \text{kV}$ or $24\ \text{kV}$ (MEA) with customer-owned step-down transformer.

3. High Voltage Customers (> 10,000 kVA / 10 MVA):
- Heavy chemical plants, steel mills, large industrial estates.
- Supplied directly from $115\ \text{kV}$ or $230\ \text{kV}$ grid via private dedicated substation.

⚠️ Exam Pitfall / Pro Tip:Large customers buy power at high voltage because tariff rates per unit (kWh) are cheaper, and transmission loss over long industrial site feeders is minimized.

CH09-07 Chapter 9 Level 1 (Core Concept) 📌 FunEE Final Note P.5 & Lecture Slides

Power Plant Specifics in Thailand (Bhumibol, Ubolratana, Mae Moh, Nam Pong)

QUESTION / EXAM PROBLEM:

Match the following major Thai power plants to their generation type, location, and key features: (1) Bhumibol Dam, (2) Ubolratana Dam, (3) Mae Moh, (4) Nam Pong.

Key Formula / Operating Principle:
Bhumibol: Ping River / Tak | Ubolratana: Khon Kaen | Mae Moh: Lignite / Lampang | Nam Pong: Combined Cycle / Khon Kaen

Step-by-Step Solution & Calculation:

1. Bhumibol Dam (เขื่อนภูมิพล):
- Location: Tak Province, Ping River.
- Thailand's first multipurpose concrete arch dam; major hydro peak-load plant.

2. Ubolratana Dam (เขื่อนอุบลรัตน์):
- Location: Khon Kaen Province, Pong River.
- First hydroelectric dam in Northeastern Thailand (Isan).

3. Mae Moh Power Plant (โรงไฟฟ้าแม่เมาะ):
- Location: Lampang Province.
- Thailand's largest coal/lignite-fired steam thermal base-load power plant ($2,625\ \text{MW}$).

4. Nam Pong Power Plant (โรงไฟฟ้าน้ำพอง):
- Location: Khon Kaen Province ($32\ \text{km}$ from downtown KKU).
- Combined-cycle power plant fueled by domestic natural gas from the Sinphuhorm/Nam Phong field.

⚠️ Exam Pitfall / Pro Tip:Very relevant for KKU engineering students! Ubolratana Dam and Nam Pong power plant are located directly in Khon Kaen.

CH09-08 Chapter 9 Level 1 (Core Concept) 📌 FunEE Final Note P.6-7 & Lecture Slides

Residential Solar PV Rooftop System Components

QUESTION / EXAM PROBLEM:

Draw the flow path and name the 5 essential components of a grid-connected residential Solar PV system.

Key Formula / Operating Principle:
PV Modules -> DC/DC Charge Controller -> Battery -> DC/AC Inverter -> Fusebox / Bi-directional Meter

Step-by-Step Solution & Calculation:

1. PV Solar Modules (Panels): Convert solar irradiance into DC electrical power.

2. DC/DC Converter / Charge Controller: Regulates charging voltage and extracts maximum power (MPPT).

3. Battery Storage: Stores electrical energy in DC form for night/emergency backup.

4. DC/AC Power Inverter: Converts direct current (DC) into synchronized $230\ \mathrm{V}, 50\ \mathrm{Hz}$ alternating current (AC).

5. Consumer Unit (Fusebox) & Bi-directional Net Meter: Distributes power to household loads and exports excess power to the MEA/PEA grid.

⚠️ Exam Pitfall / Pro Tip:Solar cells produce pure DC; household appliances run on AC. The Inverter is the crucial bridging component.

CH09-09 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.5 (Page 24 - 2 Points)

Energy Conservation in Household Washing Machines

QUESTION / EXAM PROBLEM:

State two practical, effective methods to conserve electrical energy when operating a household washing machine.

Key Formula / Operating Principle:
1. Wash full rated loads | 2. Cold water wash | 3. Air dry clothes

Step-by-Step Solution & Calculation:

1. Wash with Full Rated Load: Consolidate dirty laundry to run at maximum rated drum capacity rather than running multiple partial loads (each cycle consumes roughly the same mechanical and pumping energy).

2. Use Cold Water Wash: Avoid electrical water heating cycles. In washing machines with built-in heaters, heating water accounts for 80% to 90% of total electrical energy per wash cycle.

3. Spin and Air Dry: Utilize high-speed mechanical spin extraction and sun/air drying instead of electric heated tumble dryers.

⚠️ Exam Pitfall / Pro Tip:Direct 2-point exam question from page 24!

CH09-10 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.6 (Page 24 - 3 Points)

Energy Conservation in Industrial Manufacturing Plants

QUESTION / EXAM PROBLEM:

State three practical engineering methods to reduce electrical energy consumption and costs in an industrial manufacturing plant.

Key Formula / Operating Principle:
1. Load scheduling (TOU) | 2. Capacitor banks (PF) | 3. Transformer management

Step-by-Step Solution & Calculation:

1. Peak Load Shifting (TOU Tariff Optimization): Schedule energy-intensive machinery (electric arc furnaces, heavy crushers, pumping) during Off-Peak tariff hours (22:00 - 09:00 and weekends), dramatically lowering demand charges.

2. Power Factor Correction with Capacitor Banks: Maintain overall plant power factor above 0.95 lagging to eliminate utility reactive power penalty fees and reduce line current I^2R heat dissipation in plant distribution feeders.

3. Transformer Management & High-Efficiency Motors: Consolidate lightly loaded transformers during slow shifts to eliminate no-load core losses, and upgrade old motors to Premium Efficiency (IE3/IE4) with Variable Speed Drives (VFDs).

⚠️ Exam Pitfall / Pro Tip:Direct 3-point exam question from page 24!

PART 5

Chapter 10: Measuring Instruments & Sensors

PMMC Principles, DC Ammeters & Voltmeters with CDRX, Ayrton Shunts, Ohmmeters, Wattmeters (14 Flashcards)

CH10-01 Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.1 (Page 30 - 4 Points)

PMMC / Galvanometer Working Principle & Deflection Torque

QUESTION / EXAM PROBLEM:

Explain the working principle of a PMMC (Permanent Magnet Moving Coil) instrument. State the deflecting torque formula and deflection angle relation.

Key Formula / Operating Principle:
$T_d = BANI = BILdN$
$T_c = K\theta \implies \theta = \frac{BAN}{K} I$

Step-by-Step Solution & Calculation:

1. Principle: Motor principle (Lorentz force). When current I flows through the moving coil situated in the permanent magnetic field, a deflecting torque T_d is exerted on the coil sides.

2. Deflecting Torque: $T_d = BANI$, where B = magnetic flux density (T), A = coil area (m^2), N = turns, I = current (A).

3. Controlling Torque: Opposed by hair springs: $T_c = K\theta$.

4. Steady-State Deflection: At balance ($T_d = T_c$):

$$\theta = \left(\frac{BAN}{K}\right) I$$

Deflection θ is linearly proportional to current I, giving PMMC a perfectly UNIFORM (linear) scale.

⚠️ Exam Pitfall / Pro Tip:PMMC responds strictly to DC average current. If pure sinusoidal AC is applied, average torque is zero and pointer merely vibrates around zero.

CH10-02 Chapter 10 Level 2 (Past Exam / Calc) HW Ch.10 Problem 1 & Talui Homework Page 6

PMMC Torque & Deflection Calculation

QUESTION / EXAM PROBLEM:

A moving-coil instrument has $N = 150$ turns, coil area $A = 100\ \text{mm}^2$, air gap flux density $B = 0.15\ \text{T}$, and current $I = 5\ \text{mA}$. Spring constant $K = 2 \times 10^{-6}\ \text{N} \cdot \text{m/deg}$. Calculate: (1) Deflecting torque (T_d), (2) Deflection angle (θ in degrees and radians).

Key Formula / Operating Principle:
$T_d = NBAI$
$\theta = T_d / K$

Step-by-Step Solution & Calculation:

1. Deflecting Torque:

$A = 100\ \text{mm}^2 = 100 \times 10^{-6}\ \text{m}^2 = 10^{-4}\ \text{m}^2$

$$T_d = 150 \times 0.15 \times (100 \times 10^{-6}) \times (5 \times 10^{-3}) = 1.125 \times 10^{-5}\ \text{N} \cdot \text{m}$$

2. Deflection Angle:

$$\theta = \frac{T_d}{K} = \frac{1.125 \times 10^{-5}}{2 \times 10^{-6}} = 5.625^\circ$$

In radians:

$$\theta = 5.625 \times \frac{\pi}{180} = 0.0982\ \text{rad}$$

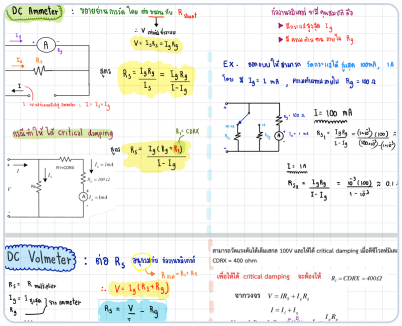
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⚠️ Exam Pitfall / Pro Tip:Always convert coil area from mm^2 to m^2 by multiplying by 10^{-6}! Never forget this conversion.

DC Ammeter Design with and without CDRX

QUESTION / EXAM PROBLEM:

Design a DC ammeter with full scale $I = 100\ \text{mA}$ using a galvanometer with $I_g = 10\ \text{mA}$ and $R_g = 50\ \Omega$. Calculate shunt resistor R_s: (a) Without damping resistor, (b) With critical damping resistor CDRX = $400\ \Omega$ connected in series with coil.



DC Ammeter with Shunt and CDRX Wiring

Key Formula / Operating Principle:

(a) $R_s = \frac{I_g R_g}{I - I_g}$

(b) $R_s = \frac{I_g(R_g + \text{CDRX})}{I - I_g}$

Step-by-Step Solution & Calculation:

(a) Without CDRX:

$$R_s = \frac{10\ \text{mA} \times 50\ \Omega}{100\ \text{mA} - 10\ \text{mA}} = \frac{0.5\ \text{V}}{0.09\ \text{A}} = 5.56\ \Omega$$

(b) With CDRX = $400\ \Omega$ (connected in series with R_g):

Total meter branch resistance $R_{\text{branch}} = R_g + \text{CDRX} = 50 + 400 = 450\ \Omega$

$$R_s = \frac{I_g(R_g + \text{CDRX})}{I - I_g} = \frac{10\ \text{mA} \times 450\ \Omega}{90\ \text{mA}} = \frac{4.5\ \text{V}}{0.09\ \text{A}} = 50.0\ \Omega$$

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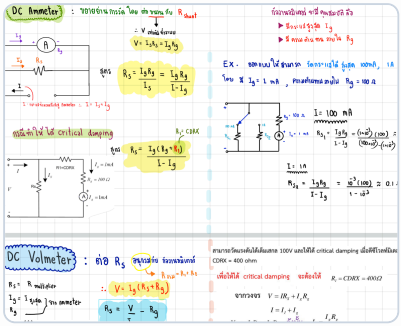
⚠️ Exam Pitfall / Pro Tip:In a DC ammeter, CDRX is connected in SERIES with the moving coil inside the meter branch; the shunt R_s is in parallel with that entire branch.

CH10-04 Chapter 10 Level 2 (Past Exam / Calc) HW Ch.10 Problem 3 & Talui Homework Page 7

DC Voltmeter Design with and without CDRX

QUESTION / EXAM PROBLEM:

Design a DC voltmeter with full scale $V = 50\ \text{V}$ using a galvanometer with $I_g = 10\ \text{mA}$ and $R_g = 50\ \Omega$. Calculate series multiplier R_s: (a) Without damping resistor, (b) With critical damping resistor CDRX = $400\ \Omega$ connected in parallel across the coil.



DC Voltmeter with Series Multiplier and Parallel CDRX

Key Formula / Operating Principle:
(a) $R_s = \frac{V}{I_g} - R_g$
(b) $R_s = \frac{V - I_g R_g}{I_g + I_{\text{cdrx}}}$

Step-by-Step Solution & Calculation:

(a) Without CDRX:

$$R_s = \frac{V}{I_g} - R_g = \frac{50}{0.010} - 50 = 5,000 - 50 = 4,950\ \Omega$$

(b) With CDRX = $400\ \Omega$ (connected in PARALLEL across the coil):

- Voltage across coil at FSD: $V_g = I_g R_g = 10\ \text{mA} \times 50\ \Omega = 0.5\ \text{V}$
- Current through CDRX: $I_{\text{cdrx}} = V_g / \text{CDRX} = 0.5/400 = 1.25\ \text{mA}$
- Total circuit current: $I_{\text{tot}} = I_g + I_{\text{cdrx}} = 10\ \text{mA} + 1.25\ \text{mA} = 11.25\ \text{mA}$
- Required series multiplier:

$$R_s = \frac{V - V_g}{I_{\text{tot}}} = \frac{50 - 0.5}{0.01125} = \frac{49.5}{0.01125} = 4,400\ \Omega$$

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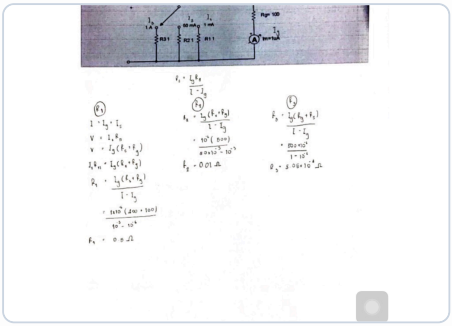
⚠️ Exam Pitfall / Pro Tip:In a voltmeter, CDRX is placed in PARALLEL with the coil to damp oscillation, so the series resistor must supply both coil current and CDRX current!

CH10-05 Chapter 10 Level 2 (Past Exam / Calc) Final Exam Problem 3.5 (Page 29 - 12 Points)

Multi-Range DC Ammeter Design with CDRX

QUESTION / EXAM PROBLEM:

Design a multi-range DC ammeter with ranges $I_1 = 1\ \text{mA}$, $I_2 = 50\ \text{mA}$, and $I_3 = 1\ \text{A}$ using a galvanometer with internal resistance $R_g = 100\ \Omega$, full-scale deflection current $I_g = 1\ \mu\text{A}$, and damping resistor $\text{CDRX} = 400\ \Omega$ in series with coil. Find shunt resistors R_1, R_2, R_3.



Exam Problem 3.5: Multi-Range DC Ammeter with CDRX Circuit

Key Formula / Operating Principle:
$R_{\text{branch}} = R_g + \text{CDRX} = 500\ \Omega$
$V_{\text{branch}} = I_g \times R_{\text{branch}} = 1\ \mu\text{A} \times 500\ \Omega = 500\ \mu\text{V} = 5 \times 10^{-4}\ \text{V}$
$R_s = \frac{V_{\text{branch}}}{I - I_g}$

Step-by-Step Solution & Calculation:
Total meter branch resistance: $R_{\text{branch}} = 100 + 400 = 500\ \Omega$.
Branch voltage at FSD: $V_m = 1 \times 10^{-6}\ \text{A} \times 500\ \Omega = 5 \times 10^{-4}\ \text{V}\ (0.5\ \text{mV})$.

1. Range 1 ($I = 1\ \text{mA} = 10^{-3}\ \text{A}$):

$$R_1 = \frac{5 \times 10^{-4}\ \text{V}}{10^{-3} - 10^{-6}} \approx \frac{5 \times 10^{-4}}{10^{-3}} = 0.5\ \Omega$$

2. Range 2 ($I = 50\ \text{mA} = 50 \times 10^{-3}\ \text{A}$):

$$R_2 = \frac{5 \times 10^{-4}\text{ V}}{50 \times 10^{-3}} = 0.01\ \Omega$$

3. Range 3 ($I = 1\ \text{A}$):

$$R_3 = \frac{5 \times 10^{-4}\ \text{V}}{1\ \text{A}} = 5 \times 10^{-4}\ \Omega = 0.0005\ \Omega = 0.5\ \text{m}\Omega$$

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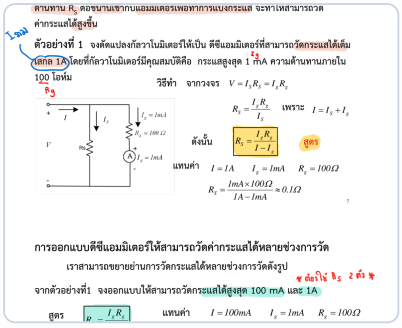
⚠️ Exam Pitfall / Pro Tip:Exact 12-point exam problem! As measurement range increases, the required shunt resistance decreases proportionally.

CH10-06 Chapter 10 Level 2 (Past Exam / Calc) FunE-final.pdf Page 18 & Slide P.12

Ayrton Shunt (Universal Shunt) for Multi-Range Ammeter

QUESTION / EXAM PROBLEM:

What is an Ayrton (Universal) Shunt in a multi-range DC ammeter? Why is it superior to individual switch shunts?



Individual Shunts vs Ayrton Universal Shunt Circuit

Key Formula / Operating Principle:
Series chain of resistors: $R_1 + R_2 + R_3$

Step-by-Step Solution & Calculation:

1. Structure: Employs a continuous series string of tapped resistors (R_1, R_2, R_3) permanently connected across the meter movement.

2. Decisive Advantage over Individual Shunts: When switching between ranges, rotary switch contacts momentarily break. With individual shunts, the meter coil is momentarily left unshunted and exposed to the full line current, burning out the delicate hair spring. With an Ayrton shunt, the meter is NEVER left open or unshunted during range switching.

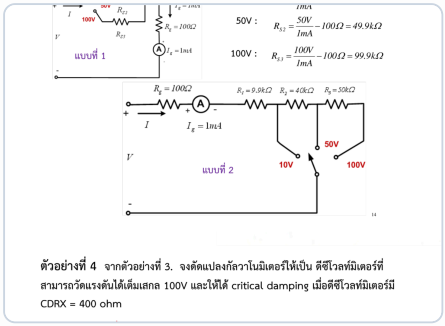
⚠️ Exam Pitfall / Pro Tip:Always mention the safety feature: the meter coil is never open-circuited while switching current ranges.

CH10-07 Chapter 10 Level 2 (Past Exam / Calc) FunE-final.pdf Page 20 & Lecture Slides

Multi-Range Voltmeter (Individual vs Tapped Series)

QUESTION / EXAM PROBLEM:

Compare Individual Multipliers versus Series Tapped Multipliers for designing a multi-range DC voltmeter. State the resistance formulas for ranges V_1, V_2, V_3.



Multi-Range Voltmeter: Individual vs Series Tapped Multipliers

Key Formula / Operating Principle:
Tapped: $R_1 = \frac{V_1}{I_g} - R_g, R_2 = \frac{V_2 - V_1}{I_g}, R_3 = \frac{V_3 - V_2}{I_g}$

Step-by-Step Solution & Calculation:

1. Method 1: Individual Multipliers (Each range uses a separate dedicated resistor):

$$R_{s1} = \frac{V_1}{I_g} - R_g, \quad R_{s2} = \frac{V_2}{I_g} - R_g, \quad R_{s3} = \frac{V_3}{I_g} - R_g$$

2. Method 2: Series Tapped Multipliers (Resistors connected in cumulative series chain):

- Range 1 (V_1): $R_1 = \frac{V_1}{I_g} - R_g$
- Range 2 (V_2): $R_2 = \frac{V_2 - V_1}{I_g}$
- Range 3 (V_3): $R_3 = \frac{V_3 - V_2}{I_g}$

Total series resistance for range 3 is $R_1 + R_2 + R_3$.

Benefit of Tapped Series: Saves board space and requires standard lower resistor values.

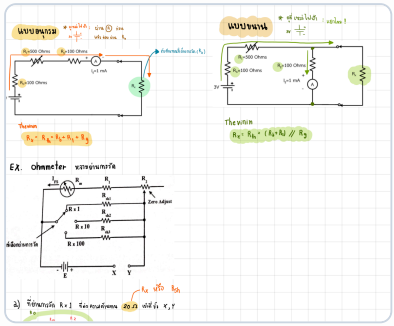
⚠️ Exam Pitfall / Pro Tip:For tapped series, remember each resistor handles only the INCREMENTAL voltage $\Delta V = V_k - V_{k-1}$, not the total voltage from zero.

CH10-08 Chapter 10 Level 1 (Core Concept) 📌 FunE-final.pdf Page 17 & Lecture Slides

Series-Type Ohmmeter Working Principle & Scale

QUESTION / EXAM PROBLEM:

Explain the operating principle of a Series-Type Ohmmeter. Describe its zero-calibration, open-circuit state, and why its scale is reversed and non-linear.



Series-Type vs Shunt-Type Ohmmeter Circuits

Key Formula / Operating Principle:

$$I_m = \frac{E}{R_1+R_2+R_g+R_x}$$

$$R_h = R_{\text{in}} = R_1 + R_2 + R_g$$

Step-by-Step Solution & Calculation:

1. Schematic: Battery E, current limiting resistor R_1, zero-adjust rheostat R_2, galvanometer R_g, and unknown resistor R_x in series.

2. Zero Calibration ($R_x = 0$, leads shorted):
Full-scale deflection current flows: $I_{\text{fsd}} = \frac{E}{R_{\text{in}}}$. Pointer deflects to FAR RIGHT ($0\ \Omega$).

3. Open Circuit ($R_x = \infty$, leads open):
Zero current flows: $I = 0$. Pointer stays at FAR LEFT ($\infty\ \Omega$).

4. Half-Scale Deflection ($I_m = 0.5I_{\text{fsd}}$):
Occurs exactly when $R_x = R_{\text{in}} = R_h$. Half-scale resistance equals the meter's internal circuit resistance.

5. Non-linear Scale: Current is inversely proportional to R_x, causing severe crowding at the left (high resistance) end.

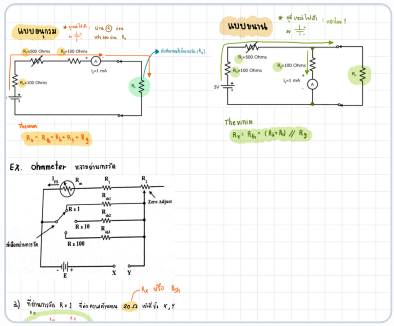
⚠️ Exam Pitfall / Pro Tip:Series ohmmeter scale runs BACKWARDS: $0\ \Omega$ is at the right, and $\infty\ \Omega$ is at the left!

CH10-09 Chapter 10 Level 1 (Core Concept) 📌 FunE-final.pdf Page 17 & Lecture Slides

Shunt-Type Ohmmeter for Low Resistance Measurement

QUESTION / EXAM PROBLEM:

Explain the operating principle of a Shunt-Type Ohmmeter. Why is its scale direction opposite to a series ohmmeter, and what resistance range is it suited for?



Shunt Ohmmeter Schematic

Key Formula / Operating Principle:
Scale: $0\ \Omega$ at LEFT, $\infty\ \Omega$ at RIGHT

Step-by-Step Solution & Calculation:

1. Structure: The unknown resistor R_x is connected in PARALLEL (shunt) across the galvanometer movement, with a switch S to disconnect the internal battery when not measuring.

2. Zero State ($R_x = 0$):
R_x completely shorts out the galvanometer coil. Meter current is zero ($I_m = 0$). Pointer stays at FAR LEFT ($0\ \Omega$).

3. Open Circuit ($R_x = \infty$):
No current is shunted away; full current flows through the meter coil ($I_m = I_{\text{fsd}}$). Pointer deflects to FAR RIGHT ($\infty\ \Omega$).

4. Application: Specifically suited for measuring very LOW resistances ($0.5\ \Omega - 100\ \Omega$), such as motor windings and transformer coils.

⚠️ Exam Pitfall / Pro Tip:Remember: Series ohmmeter has $0\ \Omega$ on RIGHT; Shunt ohmmeter has $0\ \Omega$ on LEFT (same direction as voltmeter/ammeter).

CH10-10 Chapter 10 Level 1 (Core Concept) 📌 FunE-final.pdf Page 17 & Lecture Slides

Multi-Range Ohmmeter & Zero-Adjust Rheostat Function

QUESTION / EXAM PROBLEM:

In a multi-range analog multimeter (ohmmeter), why is a Zero-Adjust Rheostat (R_{adj}) necessary, and why must the user re-zero when switching ranges ($R \times 1, R \times 10, R \times 1\text{k}$)?

Key Formula / Operating Principle:
Compensates for battery aging: $E \downarrow$ and $r_{\text{bat}} \uparrow$

Step-by-Step Solution & Calculation:

1. Battery Aging Compensation: As internal chemical batteries are used and age, their open-circuit terminal voltage gradually decreases ($E \downarrow$) and their internal resistance increases ($r_b \uparrow$). Without adjustment, full-scale deflection cannot be reached when shorting leads.

2. Zero-Adjust Rheostat Function: By turning the thumbwheel rheostat, the user varies internal resistance so that $I_{\text{fsd}} = E/R_{\text{in}}$ is precisely restored at $R_x = 0$.

3. Range Switching: Different ranges ($R \times 1, R \times 10, R \times 100$) switch in different shunt and series resistors, altering total internal circuit resistance R_{in}. Therefore, the meter MUST be re-zeroed every time a new range is selected.

⚠️ Exam Pitfall / Pro Tip:Always short the test leads and adjust to $0\ \Omega$ before making any resistance measurement!

CH10-11 Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.2 (Page 30 - 2 Points)

Rectifier AC Voltmeter & Form Factor

QUESTION / EXAM PROBLEM:

What is the operating principle of a Rectifier AC Voltmeter, and how does it relate average rectified DC to the indicated RMS value?

Key Formula / Operating Principle:
$V_{\text{rms}} = 1.11 \times V_{\text{avg}}$ (for pure sine waves)

Step-by-Step Solution & Calculation:

1. Principle: Uses semiconductor diodes (half-wave or full-wave bridge rectifier) to convert an alternating voltage into unidirectional pulsating DC, which is then measured by a standard PMMC movement.

2. Scale Calibration: PMMC inherently responds to average value ($V_{\text{avg}} = \frac{2}{\pi}V_m = 0.636V_m$). The meter scale is multiplied by the Sine Wave Form Factor:

$$\text{Form Factor} = \frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} = 1.11$$

3. Reading Distortion: If the measured waveform is not a pure sine wave (e.g. square, triangle, or inverter PWM harmonics), the reading will have significant waveform error.

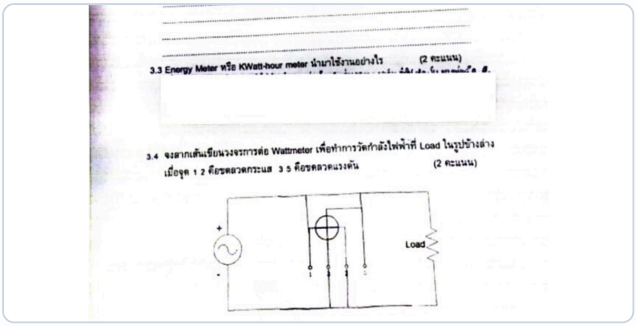
⚠️ Exam Pitfall / Pro Tip:A rectifier meter reads accurately ONLY for pure sine waves. For distorted waveforms, a True-RMS meter is required.

CH10-12 Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.4 (Page 30 - 4 Points)

Wattmeter Connection to Load (Current vs Potential Coils)

QUESTION / EXAM PROBLEM:

In an electrodynamometer Wattmeter, how must the Current Coil and Potential (Voltage) Coil be connected into the circuit to measure load power?



Exam Problem 3.4: Wattmeter Terminal Connections (1-2 and 3-5)

Key Formula / Operating Principle:
Current Coil (1-2) in SERIES | Potential Coil (3-5) in PARALLEL

Step-by-Step Solution & Calculation:

1. Current Coil (Terminals 1 - 2):

- Made of thick copper wire with very low resistance.
- Must be connected in SERIES with the load to carry full load current (I_{load}).

2. Potential / Voltage Coil (Terminals 3 - 5):

- Made of many turns of fine wire in series with a high non-inductive resistor.
- Must be connected in PARALLEL across the load to sense load voltage (V_{load}).

3. Deflection Torque:

$$T_d \propto V_{\text{load}} I_{\text{load}} \cos\theta = P \quad [\text{W}]$$

.

⚠️ Exam Pitfall / Pro Tip:Never connect the current coil in parallel across the supply—it has virtually zero resistance and will cause an explosive short circuit!

CH10-13 Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.3 (Page 30 - 2 Points)

Energy Meter (Kilowatt-Hour Meter) Function

QUESTION / EXAM PROBLEM:

What is an induction-type Energy Meter (Watt-hour meter) and what physical quantity and unit does it measure?

Key Formula / Operating Principle:
$\text{Energy} = \int P(t)dt \quad [\text{kW} \cdot \text{h}]$

Step-by-Step Solution & Calculation:

1. Purpose: An integrating instrument that measures total electrical energy consumed over a period of time, commonly used by MEA/PEA for customer billing.

2. Quantity & Unit: Measures active electrical energy in units of Kilowatt-hours ($\text{kW} \cdot \text{h}$ or 'units').

3. Principle: Eddy current induction motor principle driving a rotating aluminum disk geared to a register counter.

⚠️ Exam Pitfall / Pro Tip:A wattmeter measures instantaneous Power (W), while an energy meter measures cumulative Energy ($W \cdot h$ or $kW \cdot h$). Do not confuse the two!

Instrument Accuracy, Precision, and Error Formulas

QUESTION / EXAM PROBLEM:

Define and write the mathematical formulas for: (1) Absolute Error, (2) Relative Error (%), (3) Relative Accuracy, (4) Precision in electrical measurements.

Key Formula / Operating Principle:

$\epsilon = |Y_n - X_n|$

$\%\text{Error} = \frac{|Y_n - X_n|}{Y_n} \times 100\%$

$\%\text{Accuracy} = 100\% - \%\text{Error}$

$\text{Precision} = 1 - \left|\frac{X_n - \bar{X}_n}{\bar{X}_n}\right|$

Step-by-Step Solution & Calculation:

1. Absolute Error (ϵ): Difference between true/expected value (Y_n) and measured value (X_n):

$$\epsilon = |Y_n - X_n|$$

2. Percent Relative Error (ϵ_r):

$$\%\text{Error} = \frac{|Y_n - X_n|}{Y_n} \times 100\%$$

3. Percent Accuracy (A):

$$\%\text{Accuracy} = \left(1 - \frac{|Y_n - X_n|}{Y_n}\right) \times 100\% = 100\% - \%\text{Error}$$

4. Precision: Measure of repeatability/consistency across multiple readings:

$$\text{Precision} = 1 - \left|\frac{X_n - \bar{X}_n}{\bar{X}_n}\right|$$

where $\bar{X}_n$ is the arithmetic average of all repeated measurements.

⚠️ Exam Pitfall / Pro Tip:High precision does NOT guarantee high accuracy! A miscalibrated instrument can give highly repeatable (precise) readings that are completely wrong (inaccurate).

PART 6

Chapter 11 / Supplement: Semiconductor & Electronic Devices

Diodes, Knee Voltages, LEDs, Solar PV Cells, BJT Terminal Currents, MOSFETs, Op-Amps (12 Flashcards)

SEMI-01 Semiconductors Level 1 (Core Concept) 📌 Final Exam Problem 4.a (Page 24 - 2 Points)

Intrinsic vs Extrinsic, N-Type vs P-Type Semiconductors

QUESTION / EXAM PROBLEM:

Explain the difference between Intrinsic and Extrinsic semiconductors. Compare N-type and P-type semiconductors in terms of dopants and majority carriers.

Key Formula / Operating Principle:
N-type: Group V dopant (electrons) | P-type: Group III dopant (holes)

Step-by-Step Solution & Calculation:

1. Intrinsic Semiconductor: Chemically pure semiconductor (e.g. pure Silicon Si, Germanium Ge) with equal concentrations of free electrons and holes ($n = p = n_i$). Very low electrical conductivity at room temperature.

2. Extrinsic Semiconductor: Doped with impurity atoms to dramatically enhance conductivity:

- N-type: Pure Si doped with Pentavalent atoms (Group V: Phosphorus P, Arsenic As, Antimony Sb with 5 valence electrons). Majority carriers = Electrons; Minority carriers = Holes.
- P-type: Pure Si doped with Trivalent atoms (Group III: Boron B, Gallium Ga, Indium In with 3 valence electrons). Majority carriers = Holes; Minority carriers = Electrons.

⚠️ Exam Pitfall / Pro Tip:Exact 2-point exam question from page 24! Memory aid: P in Pentavalent gives Negative electrons; T in Trivalent gives Positive holes.

SEMI-02 Semiconductors Level 1 (Core Concept) 📌 Final Exam Problem 4.b (Page 24 - 2 Points)

MOSFET Definition, Properties & Uses

QUESTION / EXAM PROBLEM:

What is a MOSFET? State its essential electrical characteristics and its primary application/advantage in electronic circuits.

Key Formula / Operating Principle:
Metal-Oxide-Semiconductor Field-Effect Transistor (Voltage-controlled)

Step-by-Step Solution & Calculation:

1. Definition: Metal-Oxide-Semiconductor Field-Effect Transistor, a unipolar transistor where current transport involves only majority carriers.

2. Characteristics: Voltage-controlled device with extremely high input impedance ($10^{10} - 10^{15}\ \Omega$) due to the insulating SiO_2 gate oxide layer. Gate draws virtually zero DC current ($I_G \approx 0$).

3. Application / Benefit: Uses gate electric field to modulate and control current flow between Drain and Source. Widely used for high-speed switching, microprocessors (CMOS), and power converters.

⚠️ Exam Pitfall / Pro Tip:Exact question from page 24: State that it is a unipolar transistor whose output current is controlled by an electric field / voltage.

SEMI-03 Semiconductors Level 1 (Core Concept) 📌 Final Exam Problem 4.c (Page 24 - 1 Point)

Integrated Circuit (IC) Definition & Significance

QUESTION / EXAM PROBLEM:

What is an Integrated Circuit (IC)? What is its main function and significance in modern electronics?

Key Formula / Operating Principle:
Microchip integrating transistors, diodes, resistors on single silicon die

Step-by-Step Solution & Calculation:

1. Definition: A miniaturized electronic circuit consisting of thousands to billions of interconnected semiconductor devices (transistors, diodes, resistors, capacitors) fabricated on a single tiny substrate of semiconductor material (silicon wafer).

2. Significance: Enables massive reduction in physical size, weight, and power consumption, while offering dramatic improvements in reliability, operating speed, and mass-production cost.

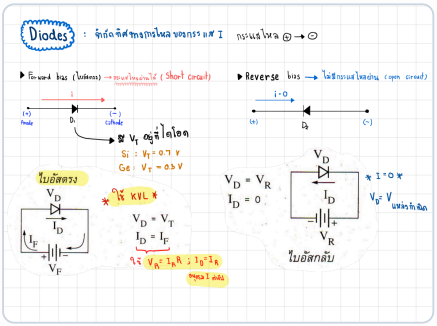
⚠️ Exam Pitfall / Pro Tip:Exact 1-point question from past exam page 24.

SEMI-04 Semiconductors Level 1 (Core Concept) 📌 Lecture Slides Ch.11 & Jele Formula Sheet

PN Junction Diode Biasing & Knee Voltages

QUESTION / EXAM PROBLEM:

Compare Forward Bias and Reverse Bias in a PN junction diode. State the barrier potential (knee voltage) values for Silicon (Si) and Germanium (Ge).



PN Diode Biasing, Forward Conduction and LED Circuit

Key Formula / Operating Principle:
Forward: P to +, N to - (conduction) | Reverse: P to -, N to + (blocking)
$V_k(\text{Si}) = 0.7\ \text{V}, V_k(\text{Ge}) = 0.3\ \text{V}$

Step-by-Step Solution & Calculation:

1. Forward Bias: External positive connected to P-side (anode), negative to N-side (cathode). Overcomes internal barrier, depletion width shrinks, and heavy forward current flows once applied voltage exceeds knee voltage.

2. Reverse Bias: Positive to N-side, negative to P-side. Depletion layer widens, blocking current except for tiny temperature-dependent reverse saturation leakage current (I_0).

3. Knee Voltages:

- Silicon (Si): $V_{\text{knee}} = 0.7\ \text{V}$
- Germanium (Ge): $V_{\text{knee}} = 0.3\ \text{V}$.

⚠️ Exam Pitfall / Pro Tip:In almost all circuit analysis, Silicon diode forward voltage drop is taken as 0.7 V.

SEMI-05 Semiconductors Level 2 (Past Exam / Calc) FunE-final.pdf Page 22 & Lecture Slides

Diode Shockley Equation, Dynamic Resistance & LED Circuit

QUESTION / EXAM PROBLEM:

1. Write the Shockley diode equation and define thermal voltage V_T at room temperature (25°C). 2. For an LED circuit with $V_{CC} = 5\ \text{V}$, LED knee voltage $V_{AK} = 2.0\ \text{V}$, find the required current limiting resistor R to maintain $I_D = 15\ \text{mA}$.

Key Formula / Operating Principle:
$I_D = I_S(e^{V_D/nV_T} - 1)$
$V_T = 25\ \text{mV}$
$R = \frac{V_{CC} - V_{AK}}{I_D}$

Step-by-Step Solution & Calculation:

1. Shockley Diode Equation:

$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

where I_S is reverse saturation current, n is ideality factor ($1 \le n \le 2$), and thermal voltage $V_T = \frac{kT}{q} \approx 25\ \text{mV}$ (or $26\ \text{mV}$) at room temperature ($298\ \text{K}$).

2. Dynamic / Small-Signal Resistance:

$$r_d = \frac{nV_T}{I_D} = \frac{25\ \text{mV}}{I_D}$$

3. LED Limiting Resistor Calculation:

$$R = \frac{V_{CC} - V_{AK}}{I_D} = \frac{5.0 - 2.0\ \text{V}}{15 \times 10^{-3}\ \text{A}} = \frac{3.0}{0.015} = 200\ \Omega$$

.

⚠️ Exam Pitfall / Pro Tip:Never connect an LED directly to a voltage supply without a series resistor; without current limiting, it will burn out instantaneously.

SEMI-06 Semiconductors 🔴 Level 2 (Past Exam / Calc) 📌 Lecture Slides Ch.11 & Jele Formula Sheet

Zener Diode Voltage Regulator Operation

QUESTION / EXAM PROBLEM:

Explain how a Zener Diode regulates voltage. A $10\ \text{V}$ Zener diode is connected across a variable input voltage $V_{\text{in}} = 15\ \text{V}$ with a series resistor $R_s = 200\ \Omega$ and load resistance $R_L = 1\ \text{k}\Omega$. Calculate Zener current I_Z.

Key Formula / Operating Principle:
$I_s = \frac{V_{\text{in}} - V_Z}{R_s}$
$I_L = V_Z / R_L$
$I_Z = I_s - I_L$

Step-by-Step Solution & Calculation:

1. Principle: Operates in the REVERSE BREAKDOWN region where voltage remains virtually constant (V_Z) across wide variations in reverse current.

2. Calculation:

- Regulated load voltage: $V_L = V_Z = 10\ \text{V}$
- Source branch current: $I_s = \frac{V_{\text{in}} - V_Z}{R_s} = \frac{15-10}{200} = \frac{5}{200} = 25\ \text{mA}$
- Load branch current: $I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{1{,}000\ \Omega} = 10\ \text{mA}$
- Zener diode current: $I_Z = I_s - I_L = 25\ \text{mA} - 10\ \text{mA} = 15\ \text{mA}$.

⚠️ Exam Pitfall / Pro Tip:For regulation to hold, I_Z must remain greater than knee current $I_{Z(min)}$ and less than maximum power rating current $I_{Z(max)} = P_Z / V_Z$.

SEMI-07 Semiconductors 🟢 Level 1 (Core Concept) 📌 FunE-final.pdf Page 22 & Lecture Slides

Solar Photovoltaic Cell Equivalent Circuit & Parameters

QUESTION / EXAM PROBLEM:

Draw the equivalent circuit of a photovoltaic solar cell and define: Open-circuit voltage (V_{oc}), Short-circuit current (I_{sc}), Maximum power point (P_{mp}), and Fill Factor (FF).

Key Formula / Operating Principle:
$\text{FF} = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}} = \frac{P_{\max}}{V_{oc} I_{sc}}$

Step-by-Step Solution & Calculation:

1. Equivalent Circuit: Light-generated constant current source I_L in parallel with a forward-biased PN junction diode D, parallel shunt leakage resistance R_{sh}, and series contact resistance R_s.

2. Key Parameters:

- V_{oc}: Open-circuit terminal voltage under illumination with zero load current (typically $0.6 - 0.7\ \text{V}$ per silicon cell).
- I_{sc}: Short-circuit current when terminals are connected together ($V = 0$).
- $P_{\max} = V_{mp} \times I_{mp}$: Point on I-V curve where power output is maximized.
- Fill Factor (FF): Ratio measuring the squareness of the I-V curve: $\text{FF} = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}}$, typically $0.75 - 0.85$ for high-quality silicon solar cells.

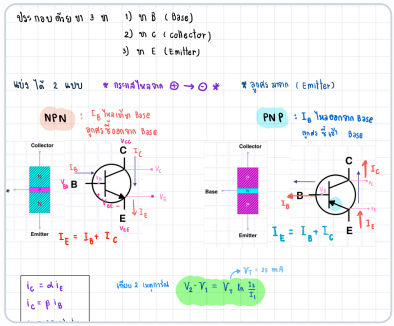
⚠️ Exam Pitfall / Pro Tip:A single silicon solar cell produces only ~0.6 V, so 60 to 72 cells are connected in series inside a panel to generate usable voltages (30-40 V).

SEMI-08 Semiconductors Level 1 (Core Concept) Final Exam Problem 4 & Lecture Slides

BJT Transistor Terminal Currents & Gain Relations

QUESTION / EXAM PROBLEM:

State the fundamental KCL current relationship between Emitter (I_e), Collector (I_c), and Base (I_b) currents in a BJT, and the relationships with current gains β and α.



BJT NPN/PNP Symbols, Terminal Currents and MOSFET Structure

Key Formula / Operating Principle:
$I_e = I_c + I_b$
$I_c = \beta I_b$
$\alpha = \frac{\beta}{\beta+1} = \frac{I_c}{I_e}$

Step-by-Step Solution & Calculation:

1. Terminal Current Law (KCL): Emitter current is the sum of collector and base currents:

$$I_e = I_c + I_b$$

2. Common-Emitter DC Current Gain (β or h_{FE}):

$$I_c = \beta I_b$$

Typically β ranges from 50 to 300.

3. Common-Base DC Current Gain (α):

$$\alpha = \frac{I_c}{I_e} = \frac{\beta}{\beta + 1}$$

Typically $\alpha \approx 0.98 - 0.998$ (slightly less than 1.0).

⚠️ Exam Pitfall / Pro Tip:Because base current I_b is tiny (typically μA), $I_e \approx I_c$ in many rough approximations, but NEVER forget $I_e = I_c + I_b$ in exact KCL equations.

SEMI-09 Semiconductors Level 1 (Core Concept) 📌 FunE-final.pdf Page 23 & Lecture Slides

BJT Transistor as an Electronic Switch

QUESTION / EXAM PROBLEM:

Explain how a BJT operates as an electronic switch. What conditions place the transistor into Cutoff (Switch OFF) versus Saturation (Switch ON)?

Key Formula / Operating Principle:
Cutoff: $I_B = 0, V_{CE} = V_{CC}$ | Saturation: $I_B \geq I_{C(\text{sat})}/\beta, V_{CE} \approx 0.2\ \text{V}$

Step-by-Step Solution & Calculation:

1. Cutoff Region (Switch OPEN / OFF):

- Condition: Base-Emitter voltage $V_{BE} < 0.7\ \text{V}$ (zero base drive).
- Currents: $I_B = 0, I_C = 0$.
- Output Voltage: $V_{CE} = V_{CC}$ (full supply voltage across open collector-emitter terminals).

2. Saturation Region (Switch CLOSED / ON):

- Condition: Base is heavily overdriven with $I_B \geq \frac{I_{C(\text{sat})}}{\beta}$, where $I_{C(\text{sat})} \approx \frac{V_{CC}}{R_C}$.
- Currents: Both junctions are forward-biased, collector current reaches maximum limited by load.
- Output Voltage: $V_{CE} = V_{CE(\text{sat})} \approx 0.2\ \text{V}$ (acts like a closed switch with minimal drop).

⚠️ Exam Pitfall / Pro Tip:When using a transistor as a switch, always ensure $I_B > I_{C(sat)}/\beta$ (overdrive factor of 2 to 3) to guarantee deep saturation.

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BJT vs FET / MOSFET Detailed Comparison

QUESTION / EXAM PROBLEM:

Provide a comprehensive comparison between BJT (Bipolar Junction Transistor) and FET / MOSFET across 6 electrical parameters.

Key Formula / Operating Principle:
BJT (Current-controlled, Bipolar) vs FET (Voltage-controlled, Unipolar)

Step-by-Step Solution & Calculation:
Direct exam comparison table from Slide 21:

1. Control Mechanism:
- BJT: Current-controlled device ($I_c = \beta I_b$).
- FET/MOSFET: Voltage-controlled device ($I_D = g_m V_{GS}$).

2. Input Impedance:
- BJT: Low ($1\ \text{k}\Omega - 10\ \text{k}\Omega$).
- FET/MOSFET: Extremely high ($10^{10} - 10^{15}\ \Omega$, draws zero gate current).

3. Charge Carriers:
- BJT: Bipolar (conduction uses both electrons and holes).
- FET: Unipolar (conduction uses only majority carriers in the channel).

4. Noise:
- BJT: More noisy.
- FET: Less noisy.

5. Thermal Stability:
- BJT: Less temperature stable (prone to thermal runaway).
- FET: More temperature stable.

6. Physical Size on Chip:
- BJT: Larger size.
- MOSFET: Much smaller size (ideal for billion-transistor IC microchips).

⚠️ Exam Pitfall / Pro Tip:Memorize this exact comparison table from Slide 21! It appears regularly in theory short-answer and multiple-choice questions.

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Half-Wave vs Full-Wave Rectifiers & PIV Rating

QUESTION / EXAM PROBLEM:

Compare Half-Wave Rectifier, Full-Wave Center-Tapped Rectifier, and Full-Wave Bridge Rectifier in terms of: (1) Number of diodes, (2) DC average output voltage (V_{dc}), (3) Peak Inverse Voltage (PIV) rating across each diode.

Key Formula / Operating Principle:
Half-wave: $V_{\text{dc}} = \frac{V_m}{\pi} = 0.318V_m$, $\text{PIV} = V_m$
Full-wave: $V_{\text{dc}} = \frac{2V_m}{\pi} = 0.636V_m$
Center-Tapped PIV = $2V_m$ | Bridge PIV = V_m

Step-by-Step Solution & Calculation:

1. Half-Wave Rectifier:

- 1 diode
- $V_{\text{dc}} = V_m/\pi = 0.318V_m$
- Ripple frequency = f_{in} ($50\ \text{Hz}$)
- $\text{PIV} = V_m$.

2. Full-Wave Center-Tapped Rectifier:

- 2 diodes
- $V_{\text{dc}} = 2V_m/\pi = 0.636V_m$
- Ripple frequency = $2f_{\text{in}}$ ($100\ \text{Hz}$)
- $\text{PIV} = 2V_m$ (Diodes must withstand twice peak voltage!).

3. Full-Wave Bridge Rectifier:

- 4 diodes
- $V_{\text{dc}} = 2V_m/\pi = 0.636V_m$
- Ripple frequency = $2f_{\text{in}}$ ($100\ \text{Hz}$)
- $\text{PIV} = V_m$ (Does NOT require expensive center-tapped transformer).

⚠️ Exam Pitfall / Pro Tip:Exam multiple choice favorite: Full-wave center-tapped diode requires $\text{PIV} = 2V_m$, whereas Bridge diode requires only $\text{PIV} = V_m$.

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Operational Amplifier (Op-Amp 741) Characteristics

QUESTION / EXAM PROBLEM:

What is an Operational Amplifier (Op-Amp)? Identify its 5 primary terminal pins and state the 3 ideal Op-Amp properties.

Key Formula / Operating Principle:
$V_{\text{out}} = A_{OL}(V_+ - V_-)$

Step-by-Step Solution & Calculation:

1. Definition: A high-gain DC-coupled differential voltage amplifier integrated circuit (e.g. standard μA741).

2. Terminal Pins:

- Pin 2: Inverting input (V_-)
- Pin 3: Non-inverting input (V_+)
- Pin 6: Output (V_{out})
- Pin 7: Positive power supply (V_S^+)
- Pin 4: Negative power supply (V_S^-).

3. Ideal Op-Amp Properties:

- Infinite open-loop voltage gain: $A_{OL} = \infty$
- Infinite input impedance: $R_{\text{in}} = \infty$ (no input current, $I_+ = I_- = 0$)
- Zero output impedance: $R_{\text{out}} = 0$
- Infinite bandwidth and zero input offset voltage.

⚠️ Exam Pitfall / Pro Tip:Virtual ground concept: When negative feedback is present, the Op-Amp drives the differential input to zero ($V_+ = V_-$).